

Convolution

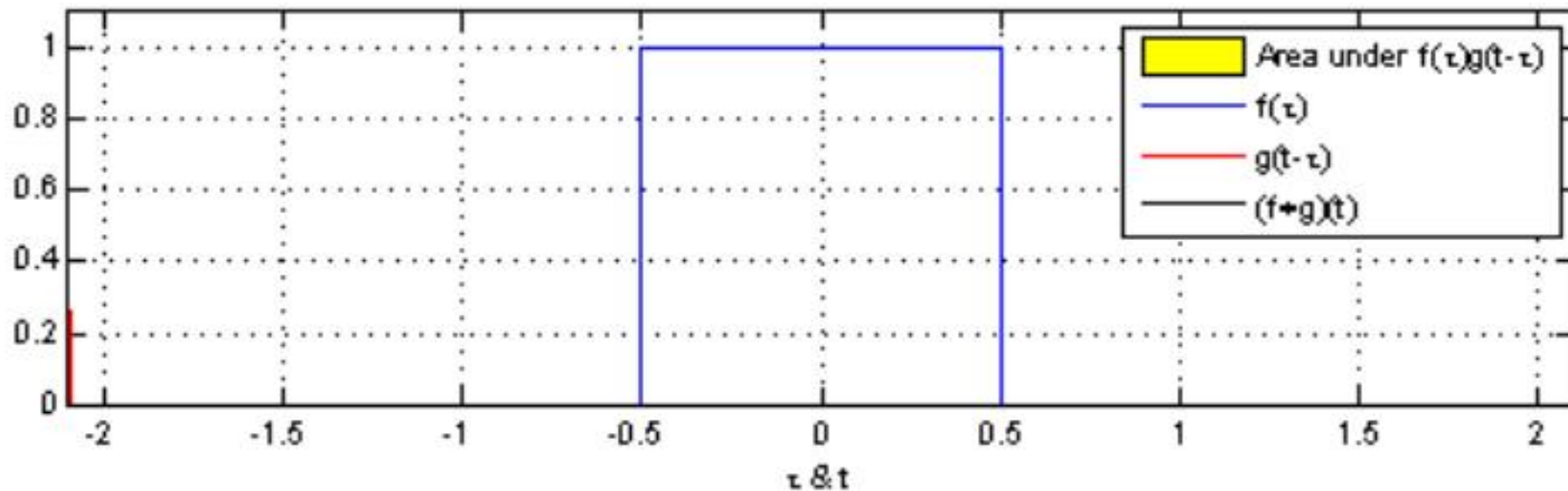
Examples and uses in images processing
Dr. Edwyn Javier Aldana Bobadilla - 2025



Introduction

Convolution

In mathematics convolution is a mathematical operation on two functions (f and g) to produce a third function that expresses how the shape of one is modified by the other.



Source: Wikipedia

Introduction

CONVOLUTION

There are several possible notations to indicate the convolution of two (multi-dimensional) signals to produce an output signal. The most common are: $\ominus$

$$c = a \otimes b = a * b$$

We shall use the first form, $c = a \otimes b$, with the following formal definitions.

Introduction

In 2D continuous space:

$$c(x, y) = a(x, y) \otimes b(x, y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} a(\chi, \zeta) b(x - \chi, y - \zeta) d\chi d\zeta$$

In 2D discrete space:

$$c[m, n] = a[m, n] \otimes b[m, n] = \sum_{j=-\infty}^{+\infty} \sum_{k=-\infty}^{+\infty} a[j, k] b[m - j, n - k]$$

Introduction



What We See

Considering only one channel

```
08 02 22 97 38 15 00 40 00 75 04 03 07 78 52 12 50 77 91 08
49 49 99 40 17 81 18 57 60 87 17 40 98 43 69 48 04 56 42 00
81 49 31 73 55 79 14 29 93 71 40 47 53 88 30 03 49 13 36 65
52 70 95 23 04 60 11 42 69 24 68 56 01 32 56 71 37 02 36 91
22 31 16 71 51 67 63 89 41 92 36 54 22 40 40 28 66 33 13 80
24 47 32 60 99 03 45 02 44 75 33 53 78 36 84 20 35 17 12 50
32 98 81 28 64 23 67 10 26 38 40 67 59 54 70 66 18 38 64 70
47 26 20 68 02 62 12 20 95 63 94 39 63 08 40 91 66 49 94 21
24 55 58 05 66 73 99 26 97 17 78 78 96 83 14 88 34 89 63 72
21 36 23 09 75 00 76 44 20 45 35 14 00 61 33 97 34 31 33 95
78 17 53 28 22 75 31 67 15 94 03 80 04 62 16 14 09 53 36 92
16 39 05 42 96 35 31 47 55 58 88 24 00 17 54 24 36 29 83 57
86 56 00 48 35 71 89 07 05 44 44 37 44 60 21 55 51 54 17 58
19 80 81 68 05 94 47 69 28 73 92 13 84 52 17 77 04 89 55 40
04 52 08 83 97 35 99 16 07 97 57 32 16 26 26 79 33 27 98 66
88 36 68 87 57 62 20 72 03 46 33 67 46 55 12 32 63 93 53 69
04 42 16 73 38 25 39 11 24 94 72 18 08 46 29 32 40 62 76 36
20 69 36 41 72 30 23 88 34 62 99 69 82 67 59 85 74 04 36 16
20 73 35 29 78 31 90 01 74 31 49 71 48 86 81 16 23 87 05 54
01 70 54 71 83 51 54 49 16 92 33 48 61 43 52 01 89 19 67 48
```

What Computers See

Introduction

Considering three channels



What We See

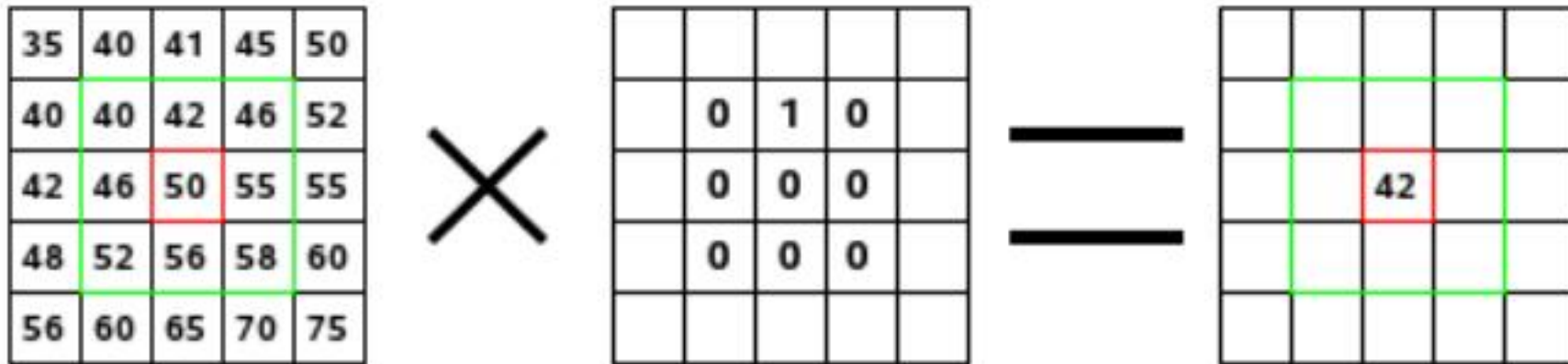
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88 36 68 87 57 62 20 72 03 46 33 67
04 42 16 73 38 25 39 11 24 94 72 18
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01 70 84 71 83 51 34 69 16 92 33 48
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What Computers See

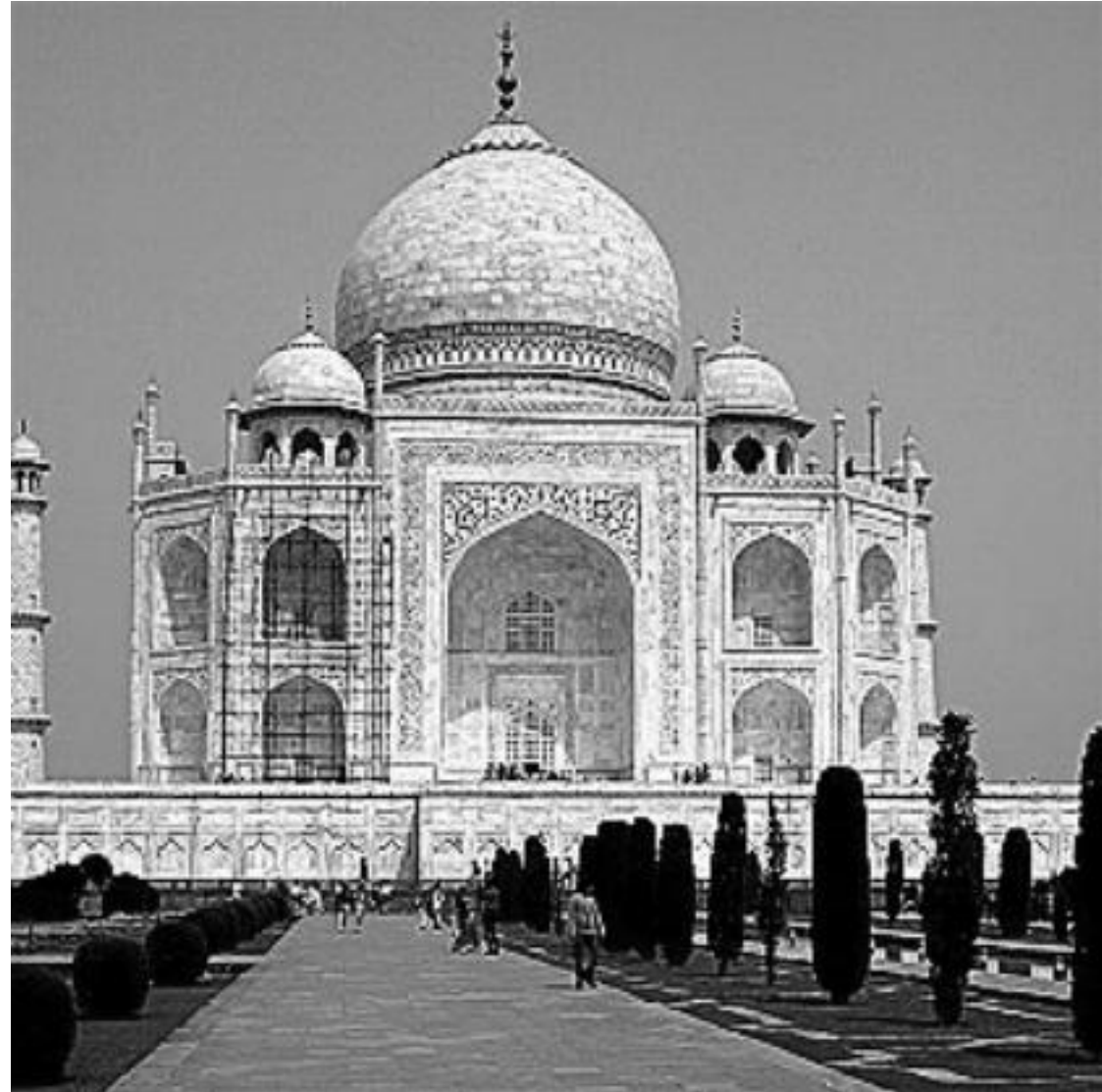
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01 70 84 71 83 51 34 69 16 92 33 48
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08 02 22 97 38 15 00 40 00 75 04 05 07 78 52 12 50 77 91 08
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88 36 68 87 57 62 20 72 03 46 33 67
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Introduction

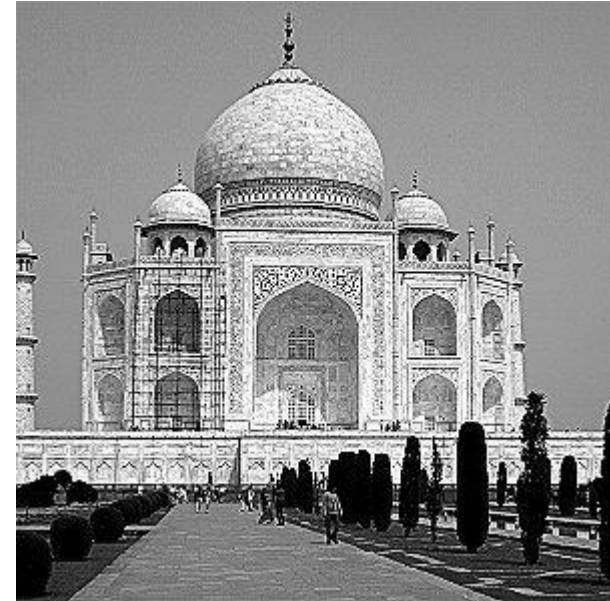
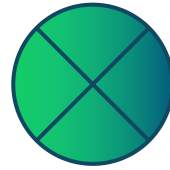


Introduction

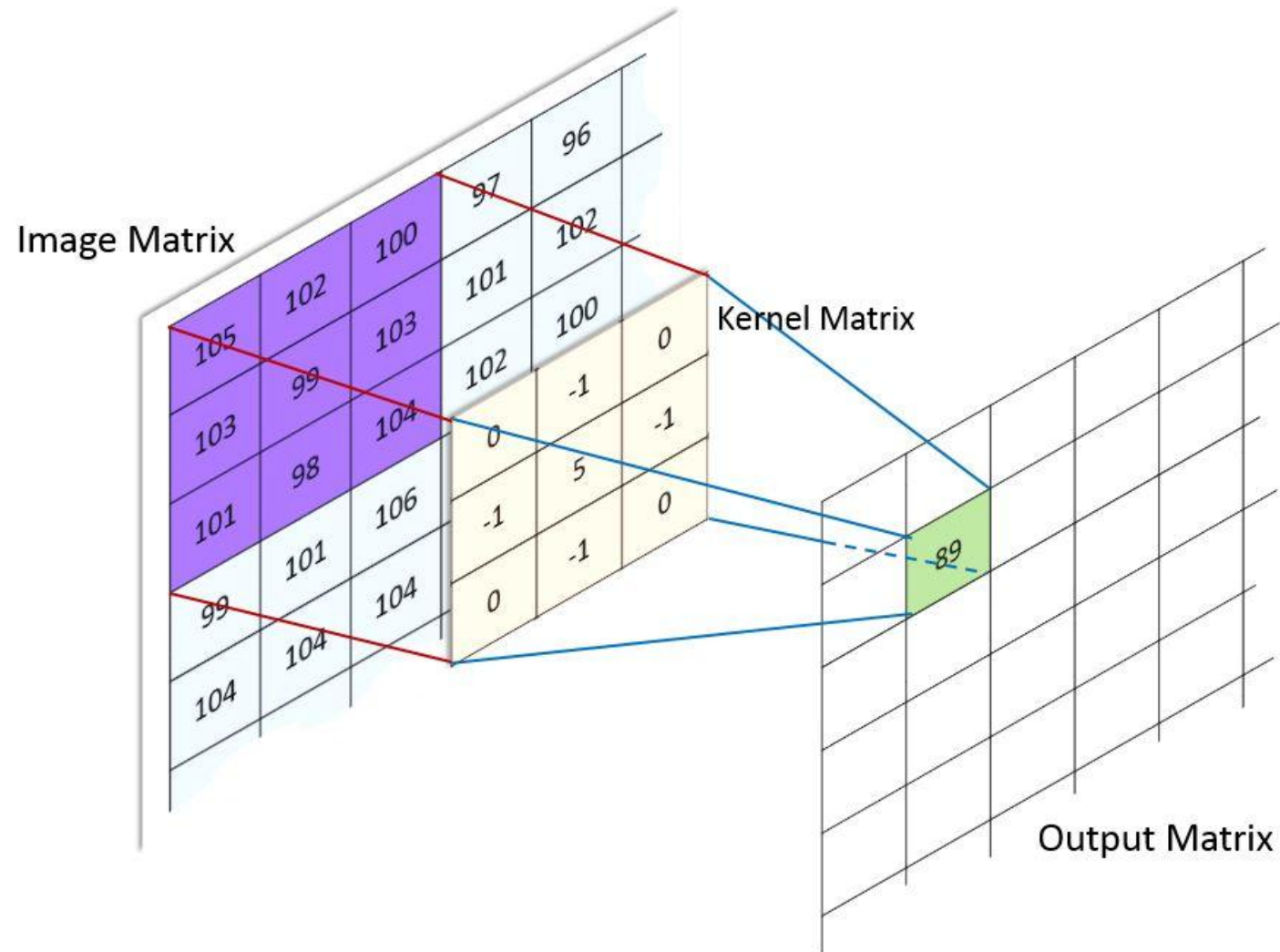


Introduction

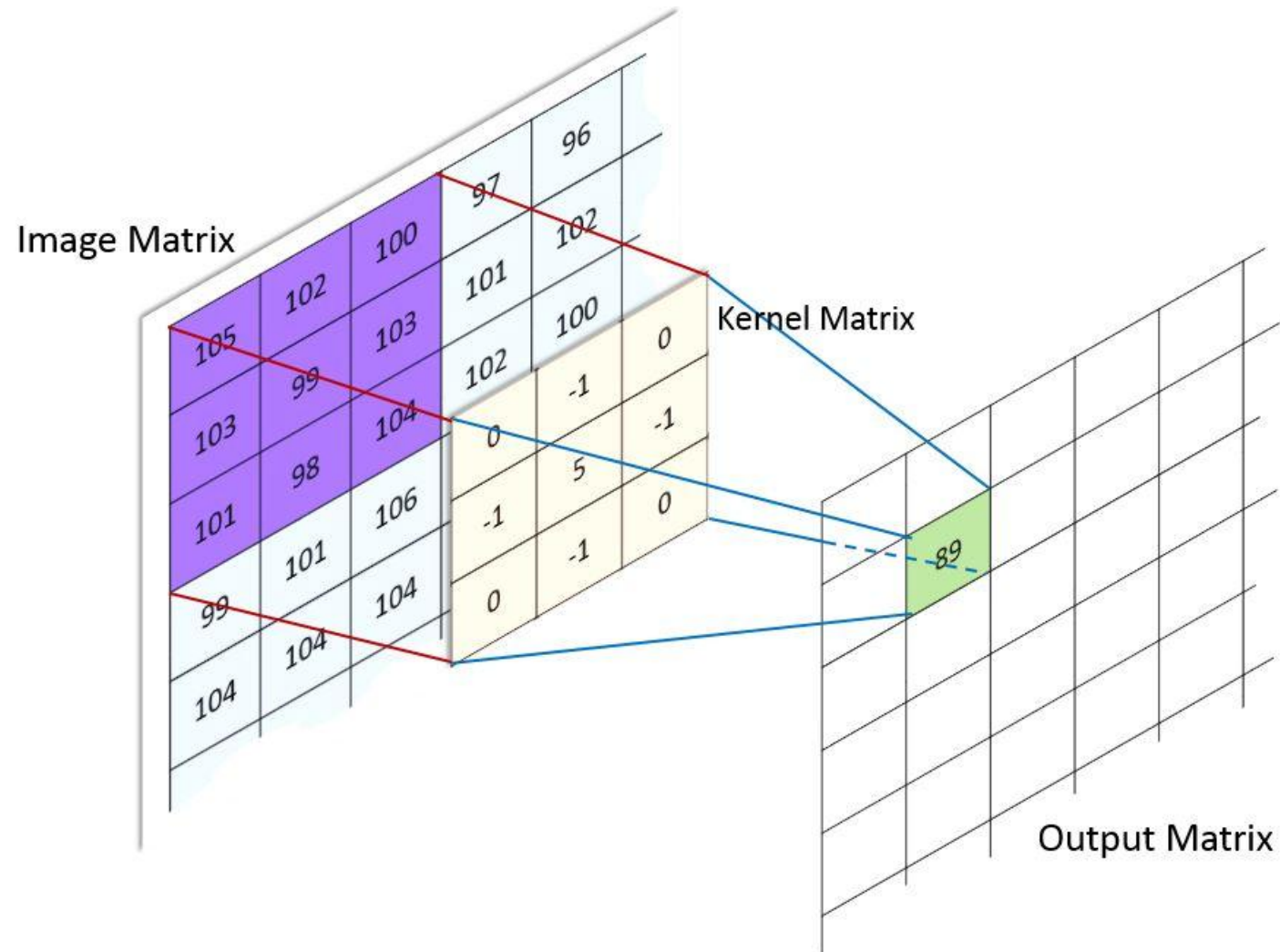
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0	0	-1	0	0
0	-1	5	-1	0
0	0	-1	0	0
0	0	0	0	0



Introduction



Introduction



Introduction

0	0	0	0	0
0	0	-1	0	0
0	-1	5	-1	0
0	0	-1	0	0
0	0	0	0	0



Introduction

1 _{x1}	1 _{x0}	1 _{x1}	0	0
0 _{x0}	1 _{x1}	1 _{x0}	1	0
0 _{x1}	0 _{x0}	1 _{x1}	1	1
0	0	1	1	0
0	1	1	0	0

Image

4		

Convolved
Feature

Kernel Effects

input image

$$\begin{pmatrix} 125 & 133 & 131 \\ \times -2 & \times -1 & \times 0 \\ + 154 & 135 & 130 \\ \times -1 & \times 1 & \times 1 \\ + ? & ? & ? \\ \times 0 & \times 1 & \times 2 \end{pmatrix} = ?$$

kernel:

output image

input image

$$\begin{pmatrix} 151 & 147 & 144 \\ \times 1 & \times 2 & \times 1 \\ + 141 & 144 & 148 \\ \times 0 & \times 0 & \times 0 \\ + 143 & 142 & 141 \\ \times -1 & \times -2 & \times -1 \end{pmatrix} = 21$$

kernel:

output image

input image

$$\begin{pmatrix} 76 & 34 & 33 \\ \times 0.0625 & \times 0.125 & \times 0.0625 \\ + 35 & 30 & 23 \\ \times 0.125 & \times 0.25 & \times 0.125 \\ + ? & ? & ? \\ \times 0.0625 & \times 0.125 & \times 0.0625 \end{pmatrix} = ?$$

kernel:

output image

input image

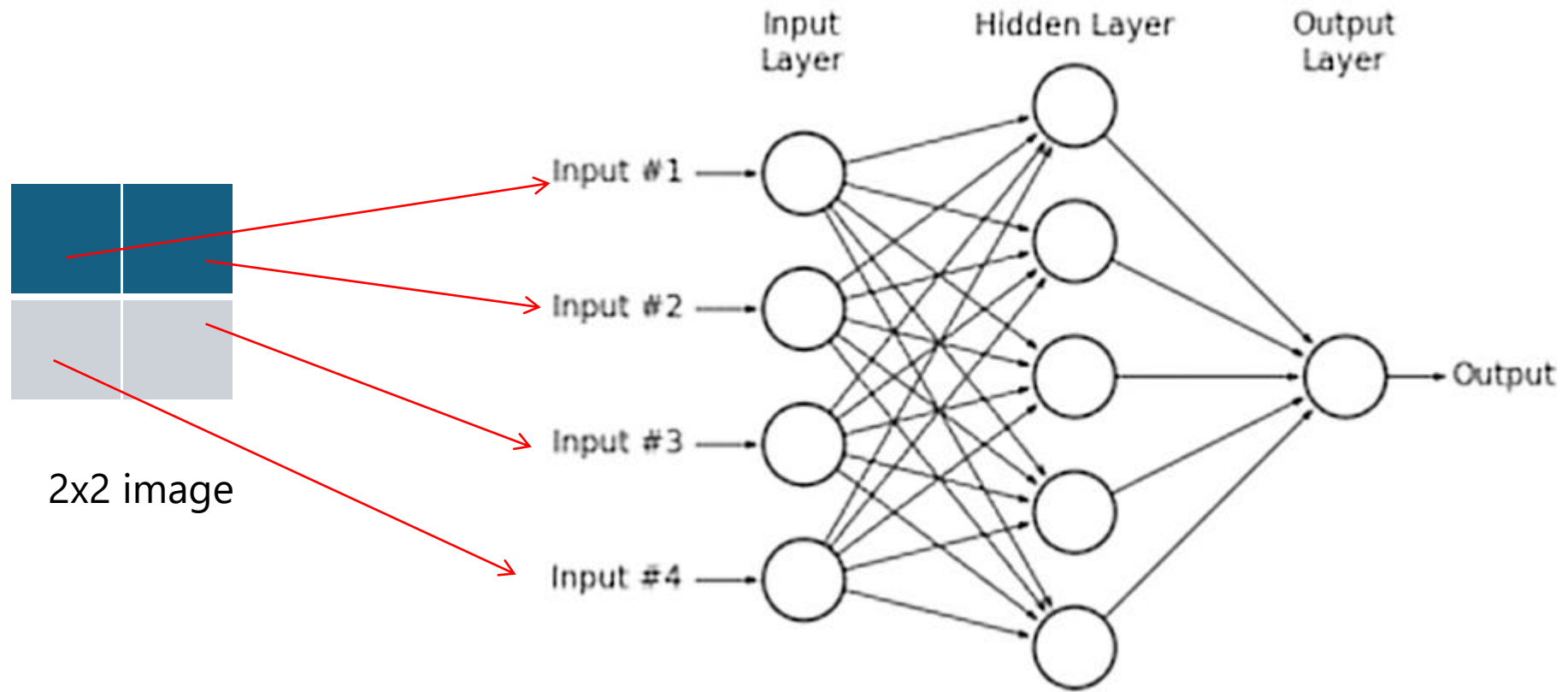
$$\begin{pmatrix} 133 & 131 & 131 \\ \times 1 & \times 0 & \times -1 \\ + 135 & 130 & 129 \\ \times 2 & \times 0 & \times -2 \\ + ? & ? & ? \\ \times 1 & \times 0 & \times -1 \end{pmatrix} = ?$$

kernel:

output image

see: <https://setosa.io/ev/image-kernels/>

Introduction



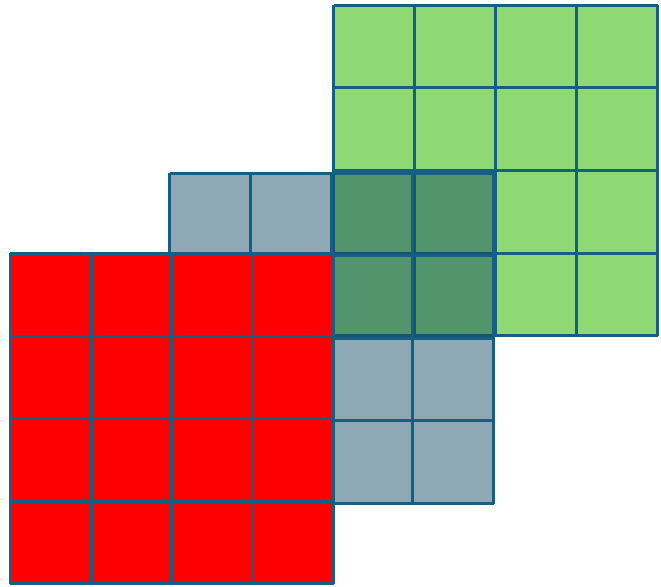
Exercise

- Download the script *Deep_Intro_Convolution* and visualize the effect of different kernels on the image cat.jpg

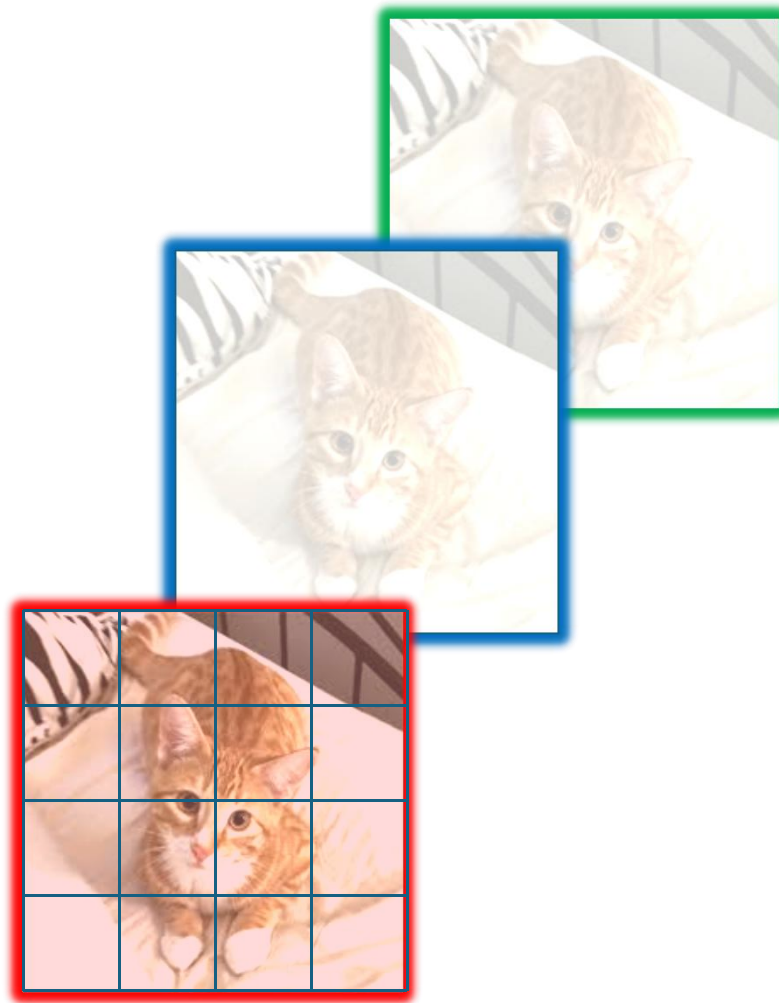
Convolutional Neural Networks (CNNs)

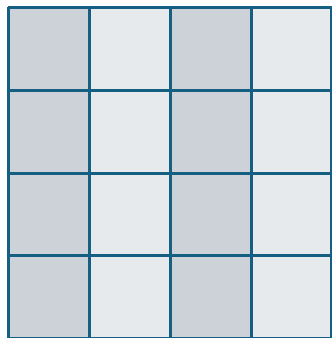
Deep Learning Foundations
Dr. Edwyn Javier Aldana Bobadilla - 2025





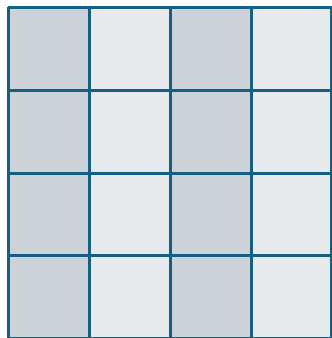
4x4
Three channels



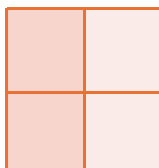
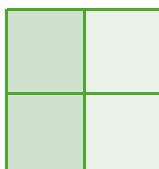
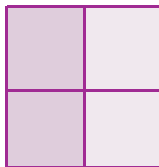
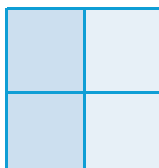


4x4

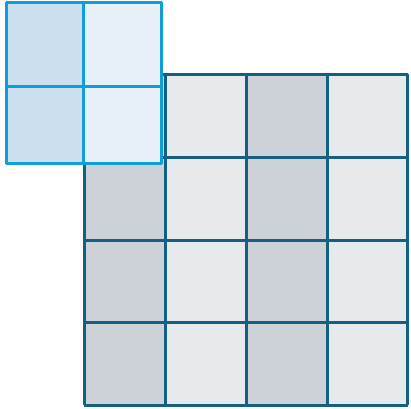
One channel



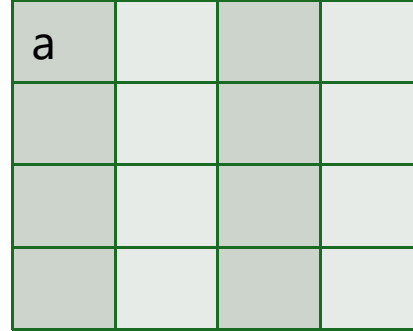
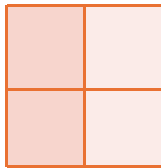
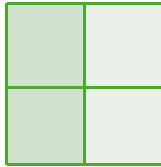
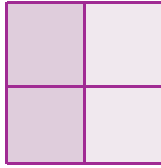
4x4
One channel



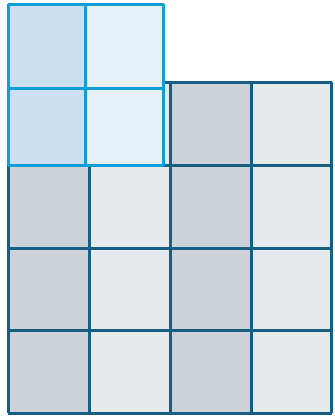
Convolution Stage



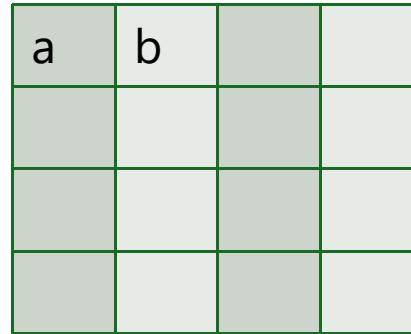
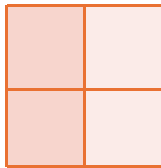
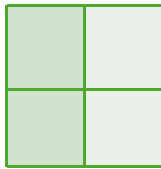
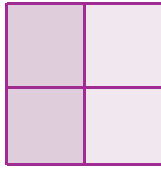
4x4
One channel



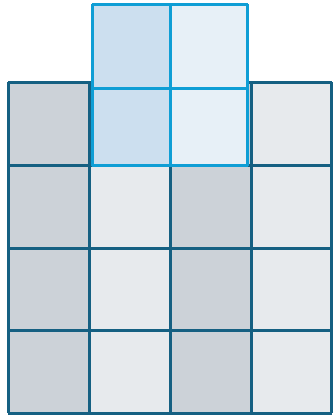
Convolution Stage



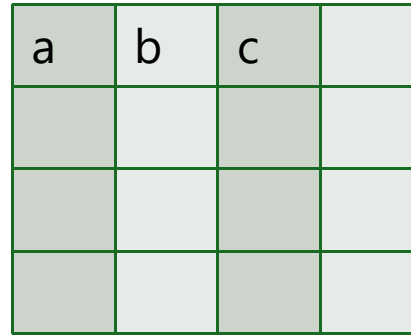
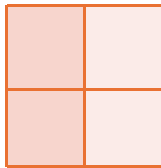
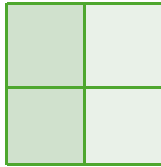
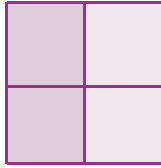
4x4
One channel



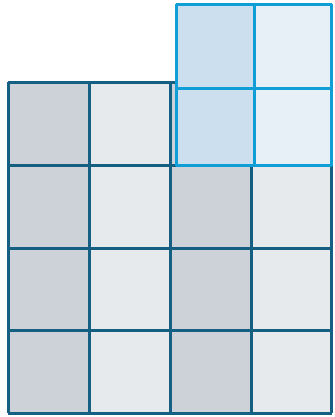
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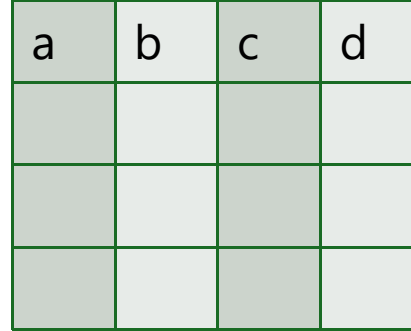
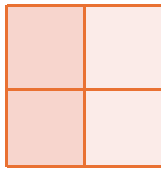
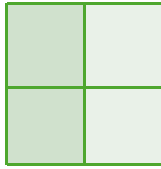
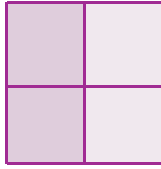
4x4
One channel



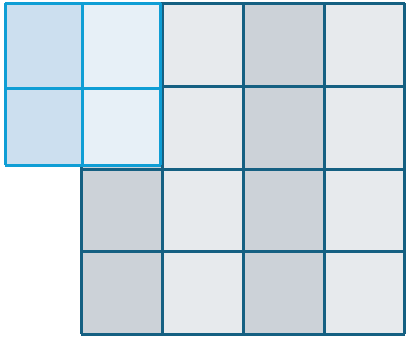
Convolution Stage



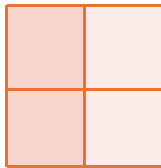
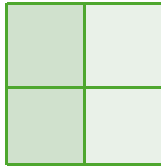
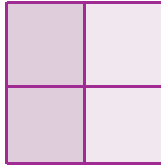
4x4
One channel



Convolution Stage

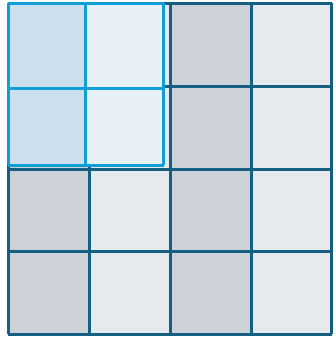


4x4
One channel

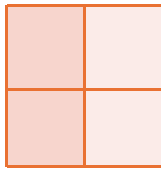
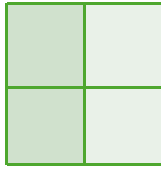
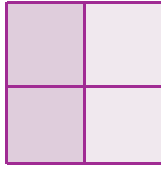


a	b	c	d
e			

Convolution Stage

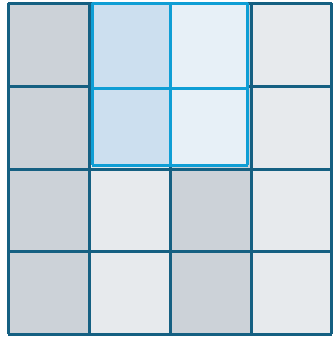


4x4
One channel

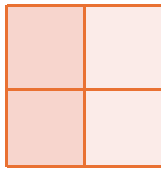
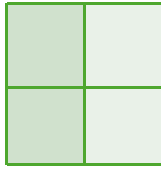
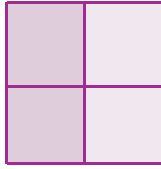


a	b	c	d
e	f		

Convolution Stage

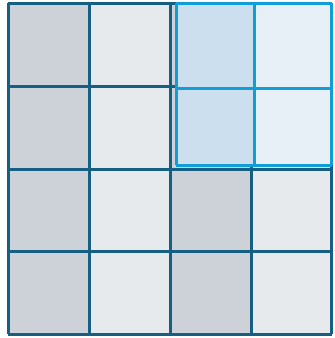


4x4
One channel

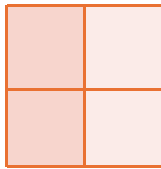
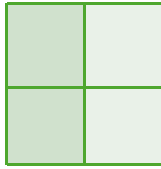
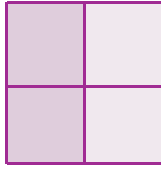


a	b	c	d
e	f	g	

Convolution Stage

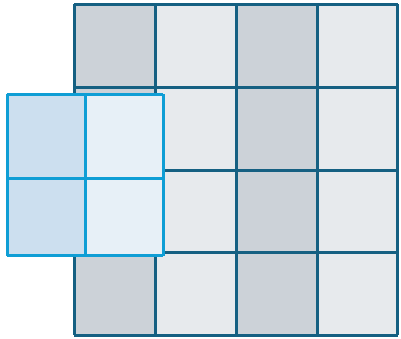


4x4
One channel

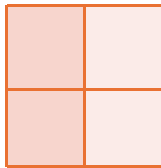
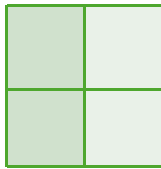
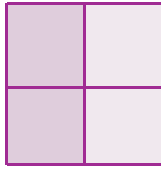


a	b	c	d
e	f	g	h

Convolution Stage

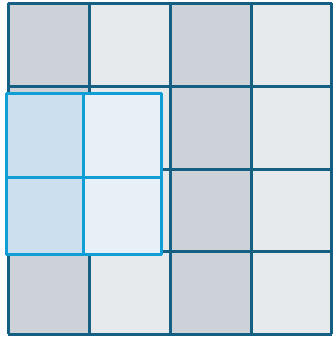


4x4
One channel

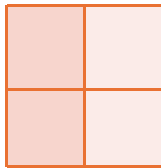
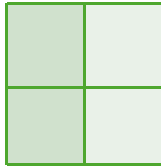
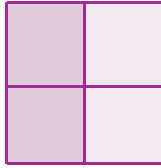


a	b	c	d
e	f	g	h
i			

Convolution Stage

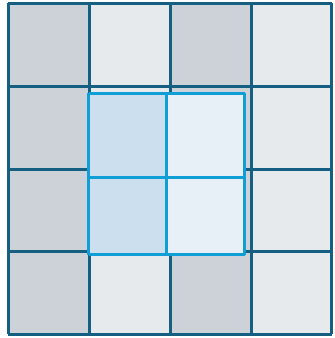


4x4
One channel

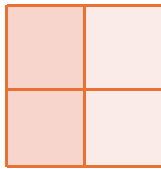
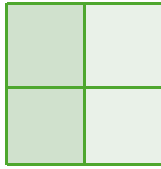
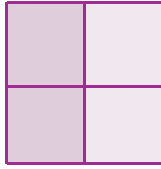


a	b	c	d
e	f	g	h
i	j		

Convolution Stage

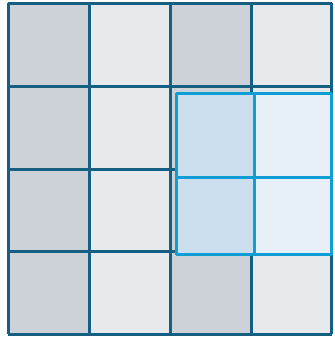


4x4
One channel

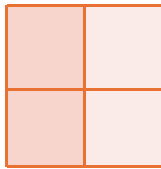
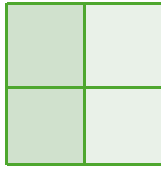
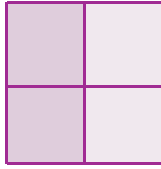


a	b	c	d
e	f	g	h
i	j	k	

Convolution Stage

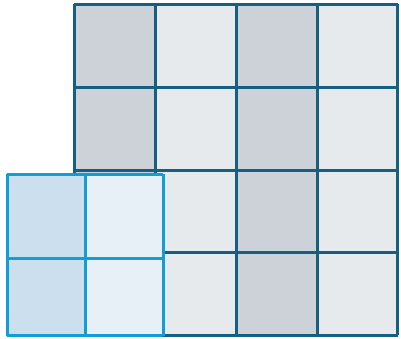


4x4
One channel

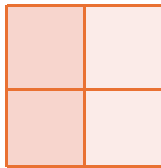
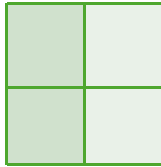
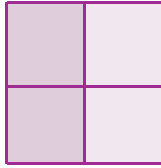


a	b	c	d
e	f	g	h
i	j	k	l

Convolution Stage

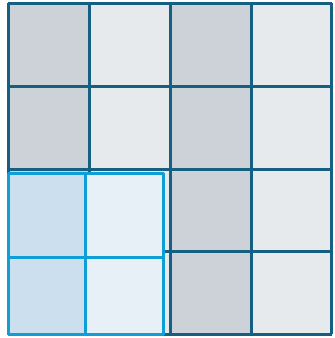


4x4
One channel

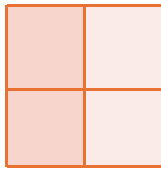
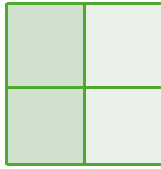
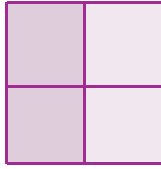


a	b	c	d
e	f	g	h
i	j	k	l
m			

Convolution Stage

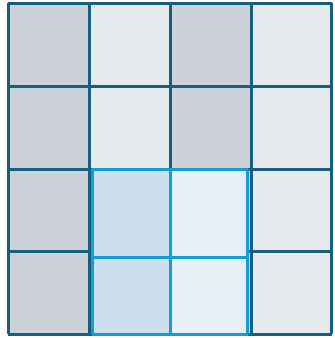


4x4
One channel

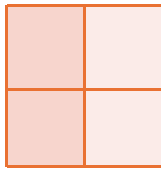
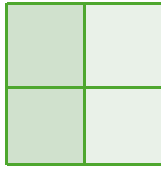
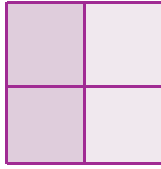


a	b	c	d
e	f	g	h
i	j	k	l
m	n		

Convolution Stage

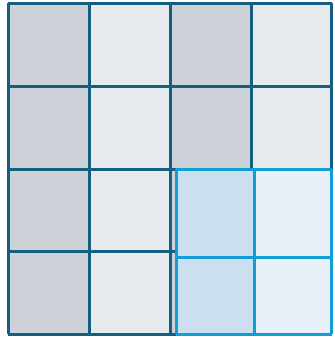


4x4
One channel

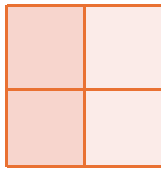
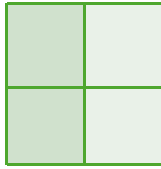
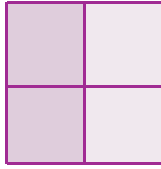


a	b	c	d
e	f	g	h
i	j	k	l
m	n	l	

Convolution Stage

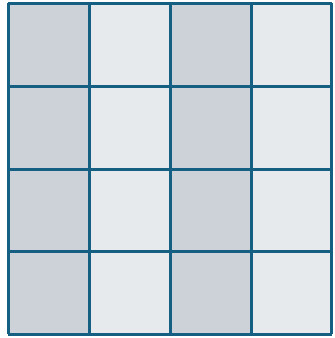


4x4
One channel

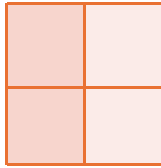
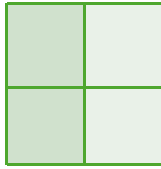
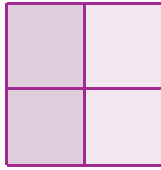


a	b	c	d
e	f	g	h
i	j	k	l
m	n	l	o

Convolution Stage

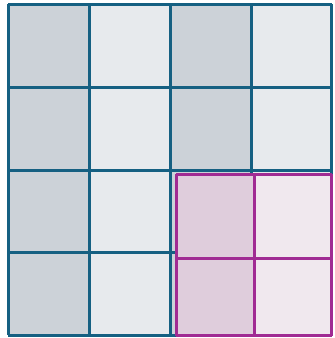


4x4
One channel

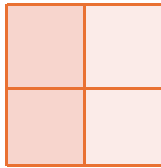
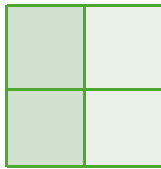


a	b	c	d
e	f	g	h
i	j	k	l
m	n	l	o

Convolution Stage

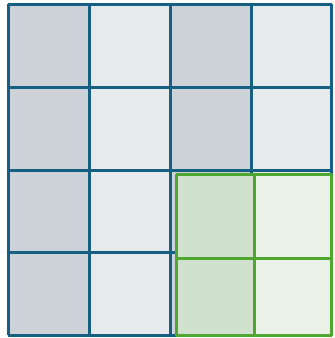


4x4
One channel

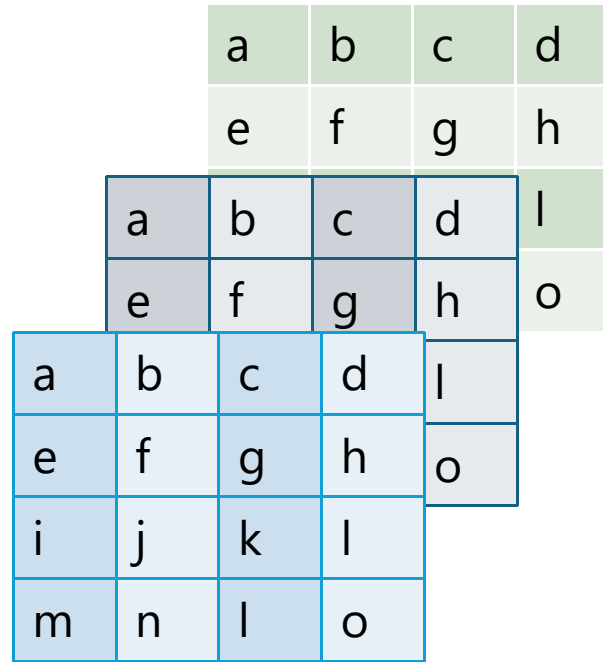
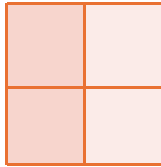


	a	b	c	d
	e	f	g	h
a	b	c	d	l
e	f	g	h	o
i	j	k	l	
m	n	l	o	

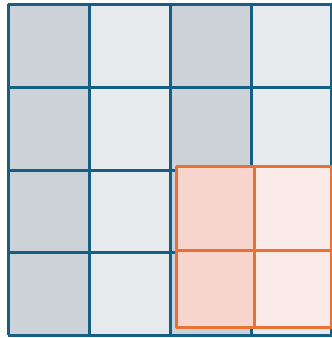
Convolution Stage



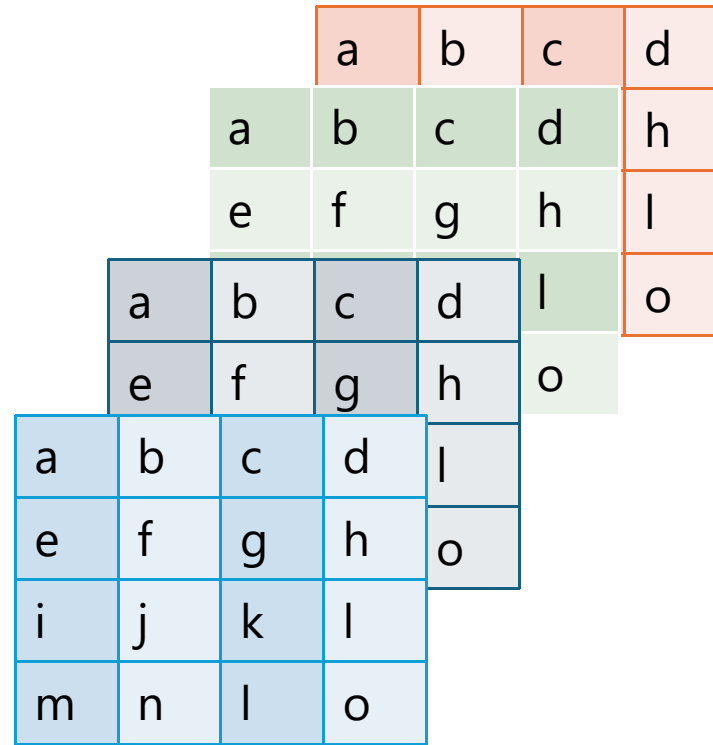
4x4
One channel



Convolution Stage

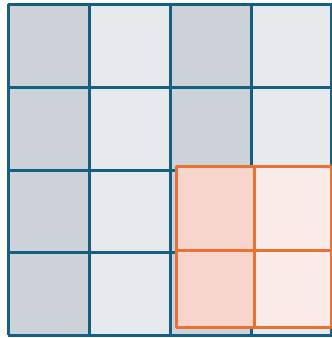


4x4
One channel

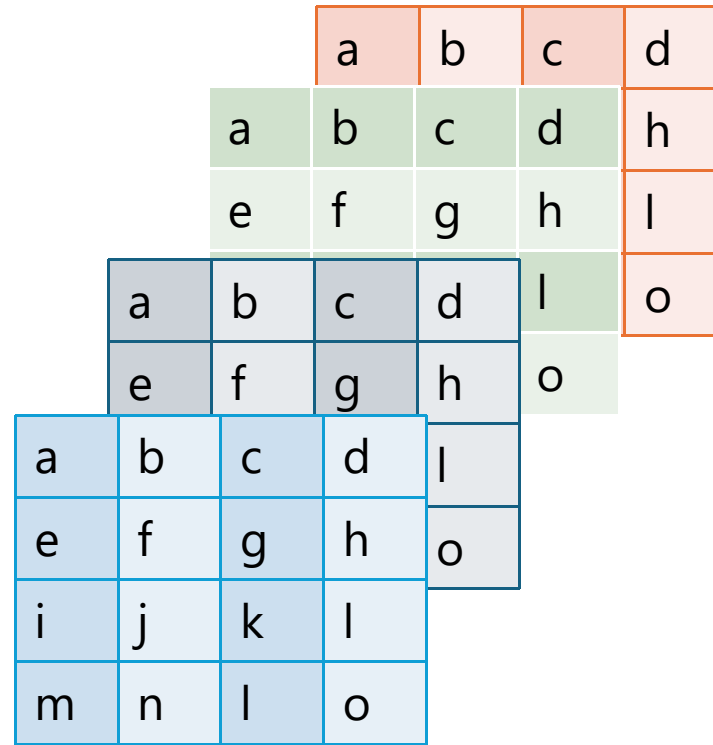


4X4x4

Convolution Stage

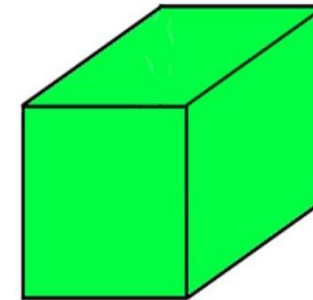


4x4
One channel



4X4x4

3D TENSOR/
CUBE

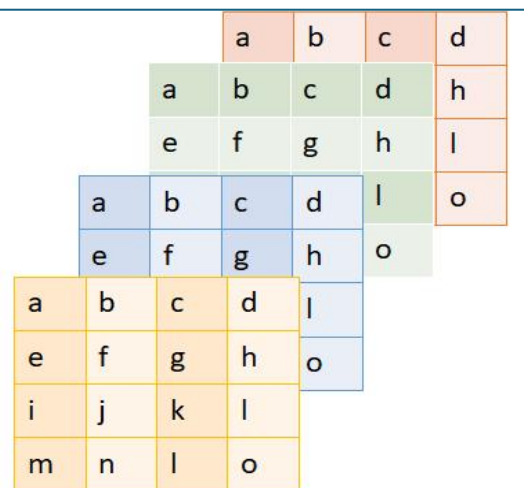


-9	4	2	5	7
3	0	12	8	61
1	23	-6	45	2
22	3	-1	72	6

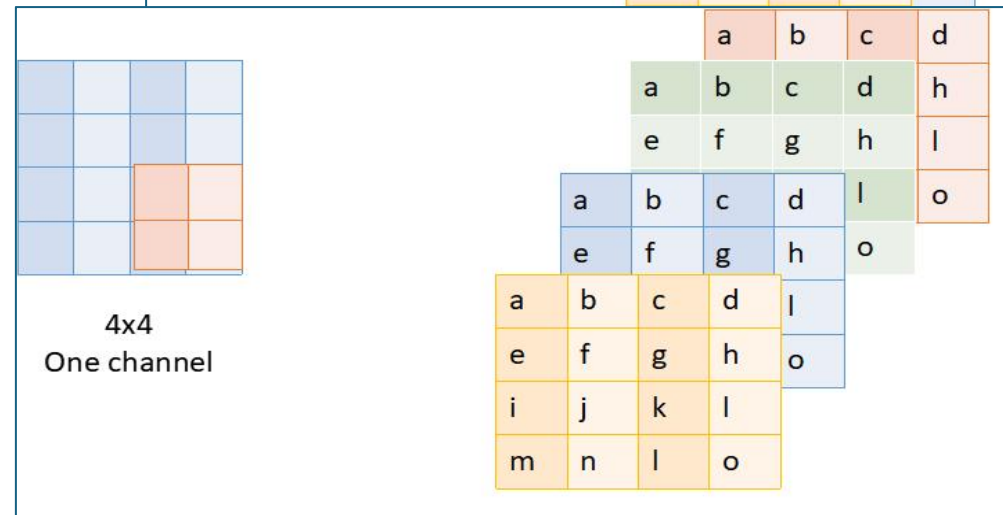
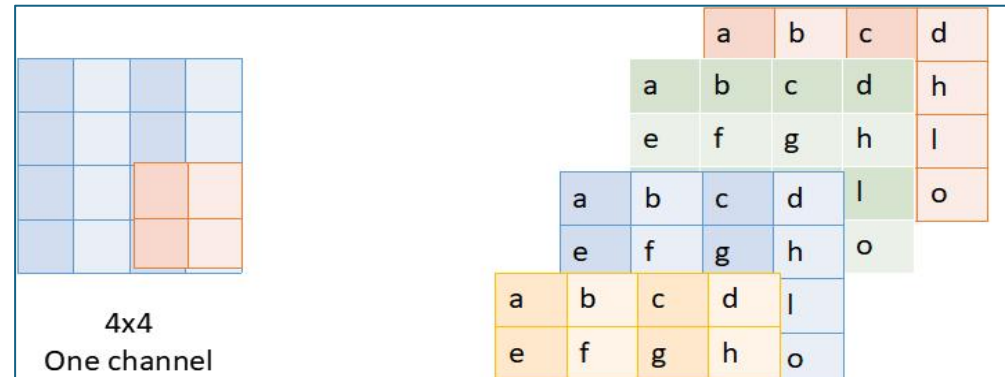
Convolution Stage



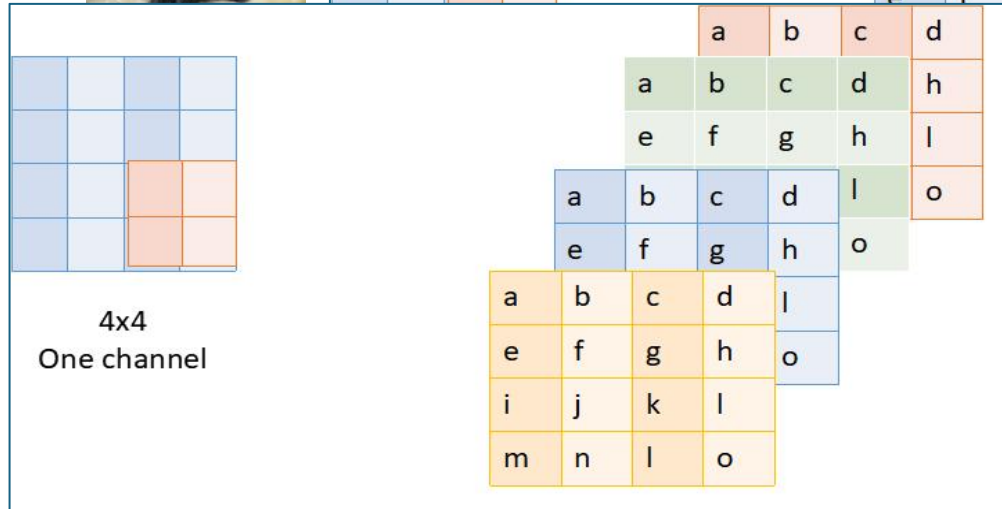
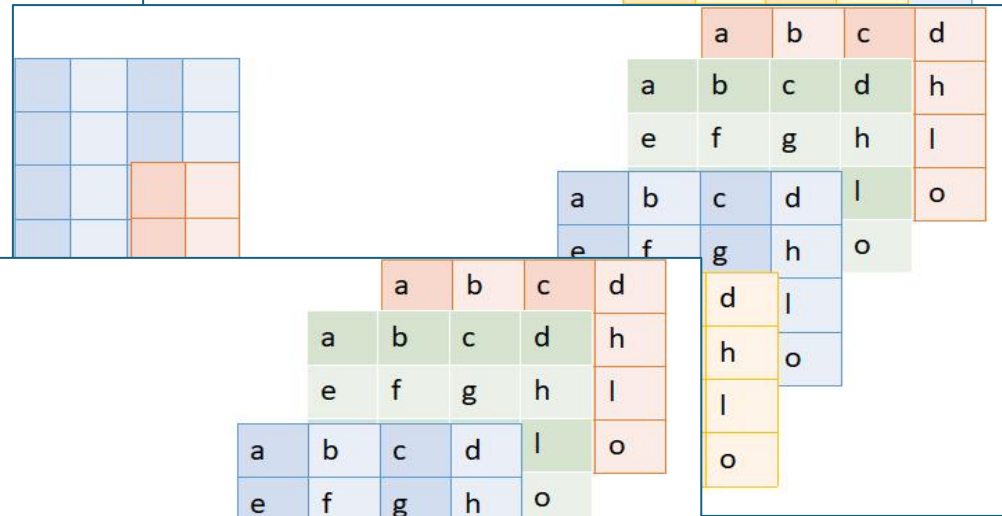
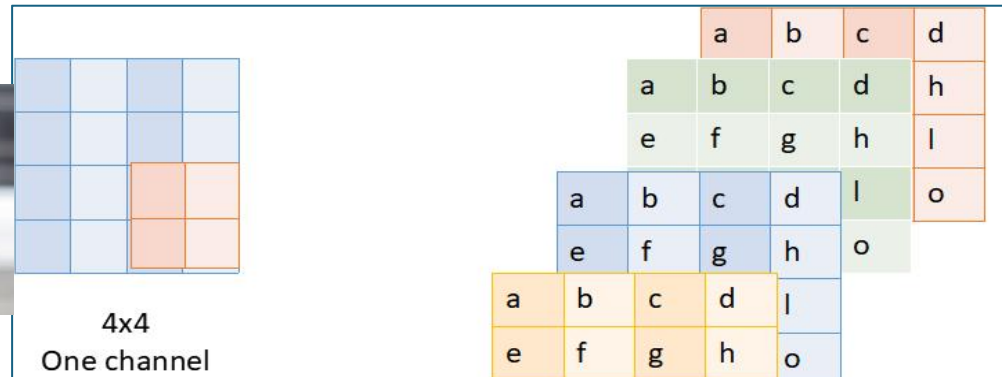
4x4
One channel



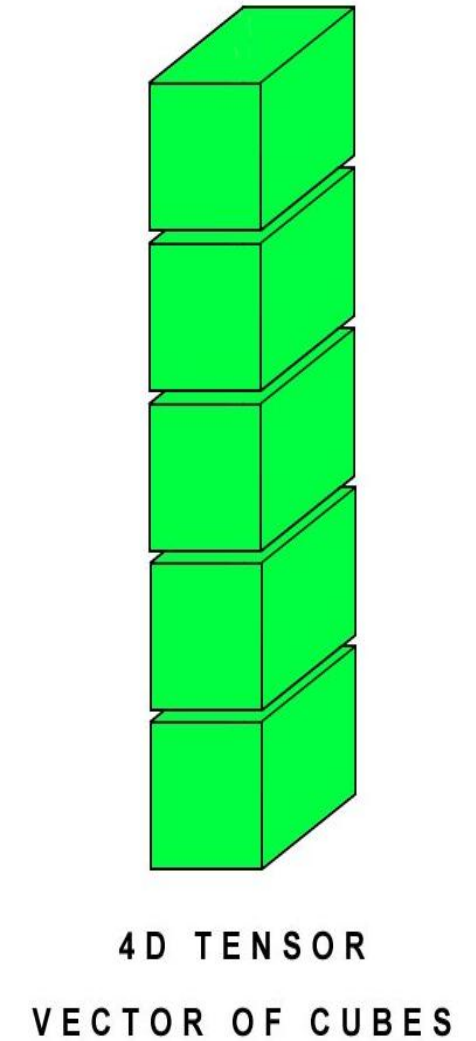
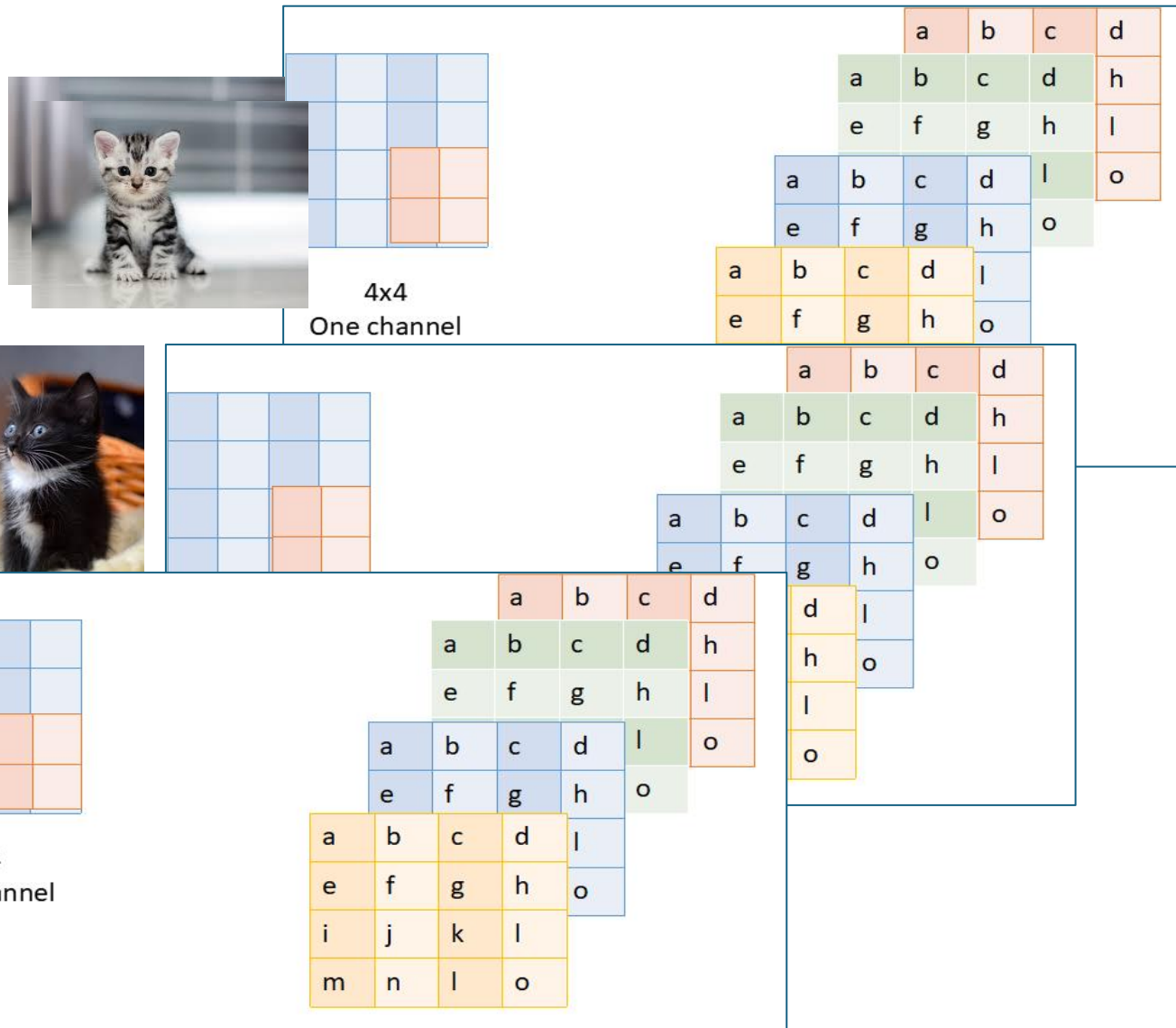
Convolution Stage



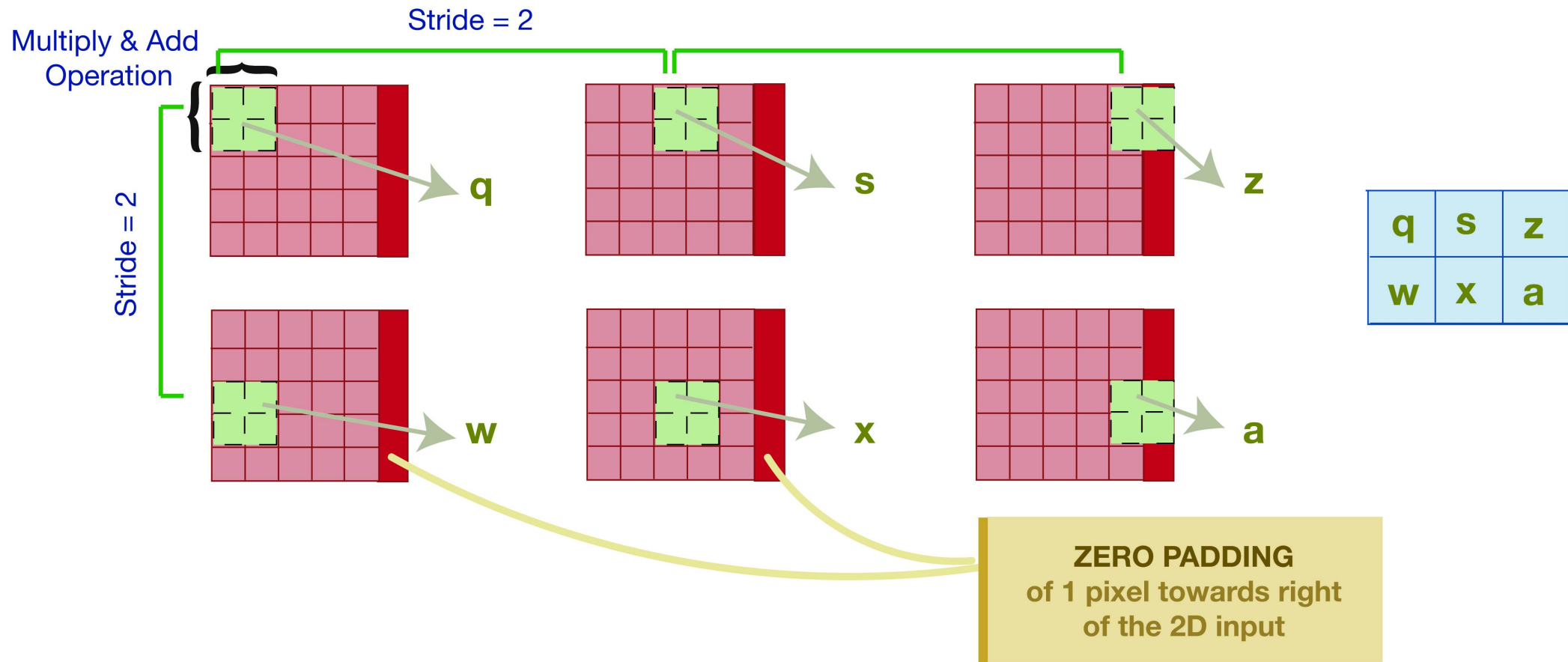
Convolution Stage



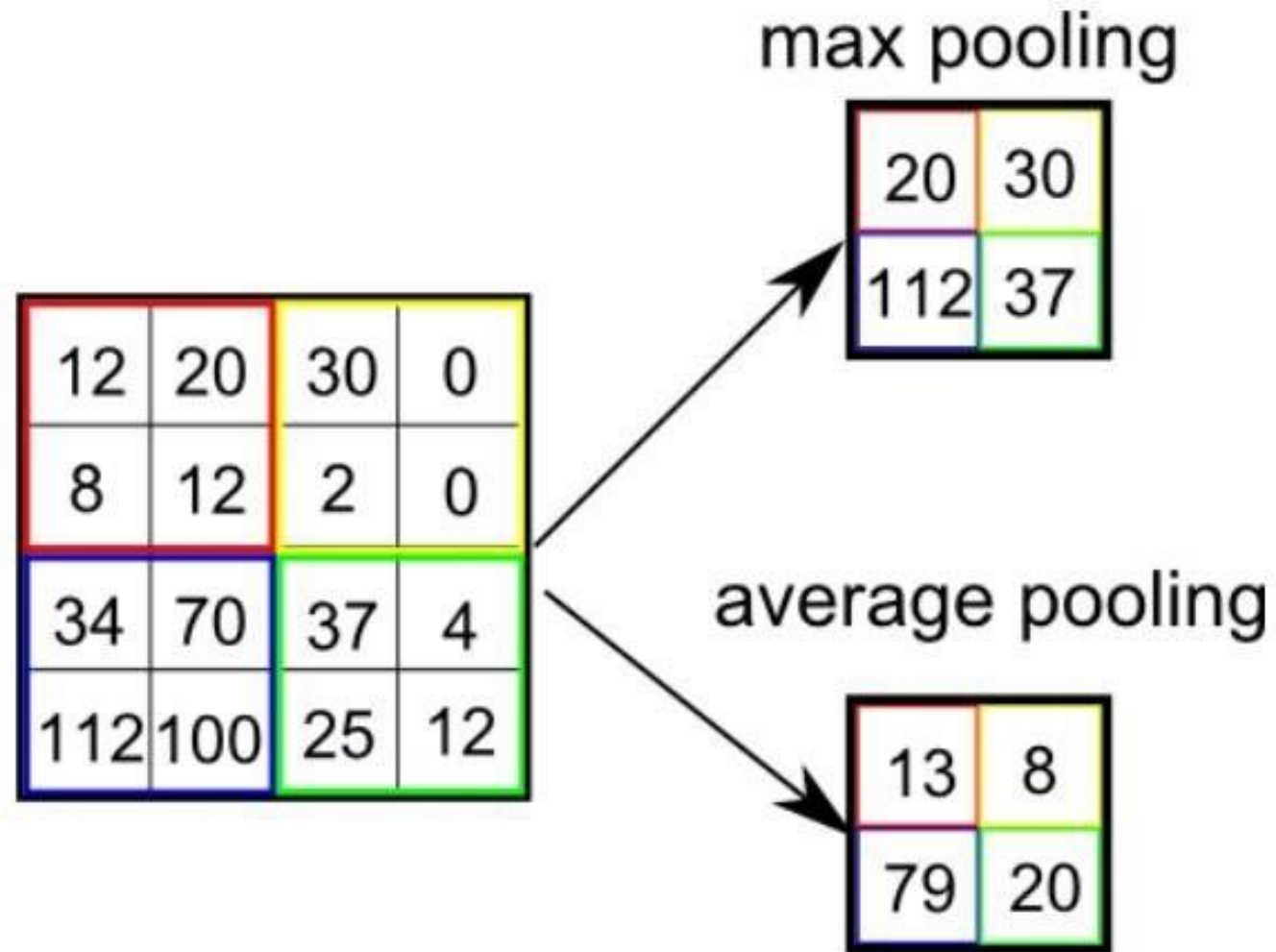
Convolution Stage



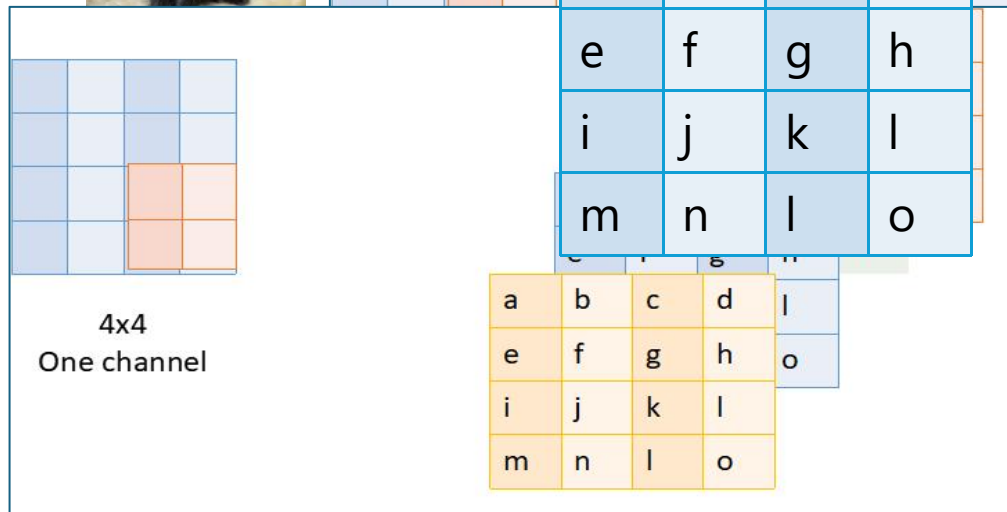
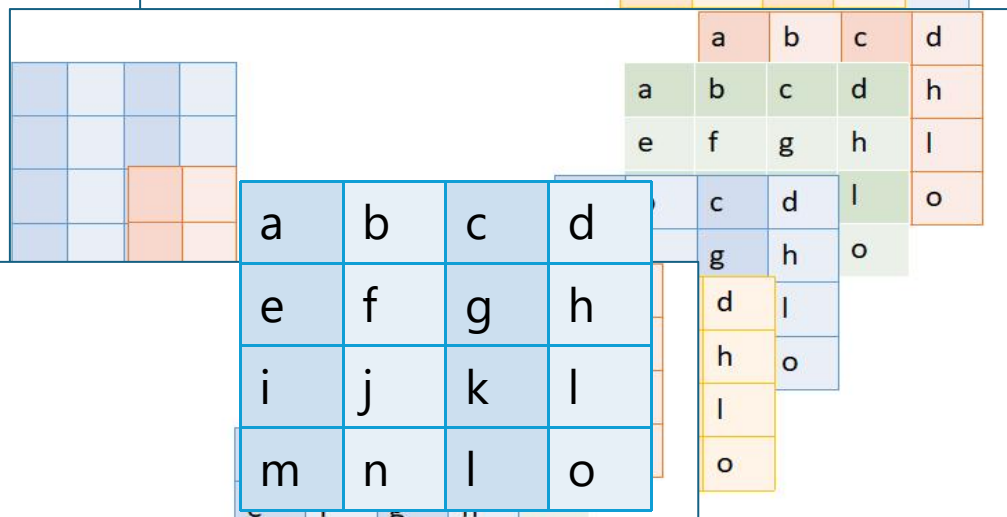
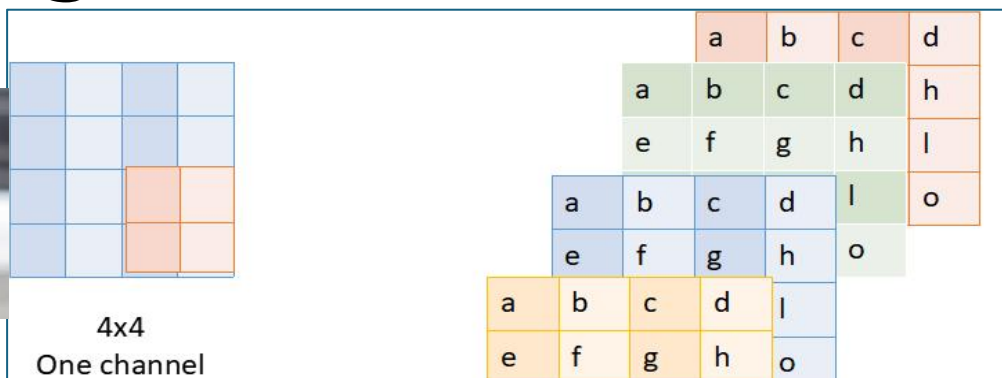
Stride and Padding



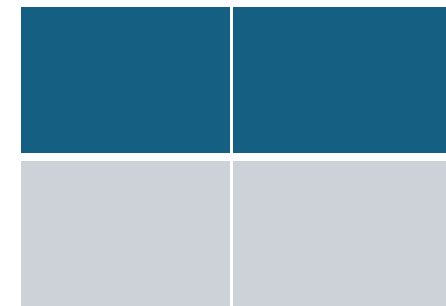
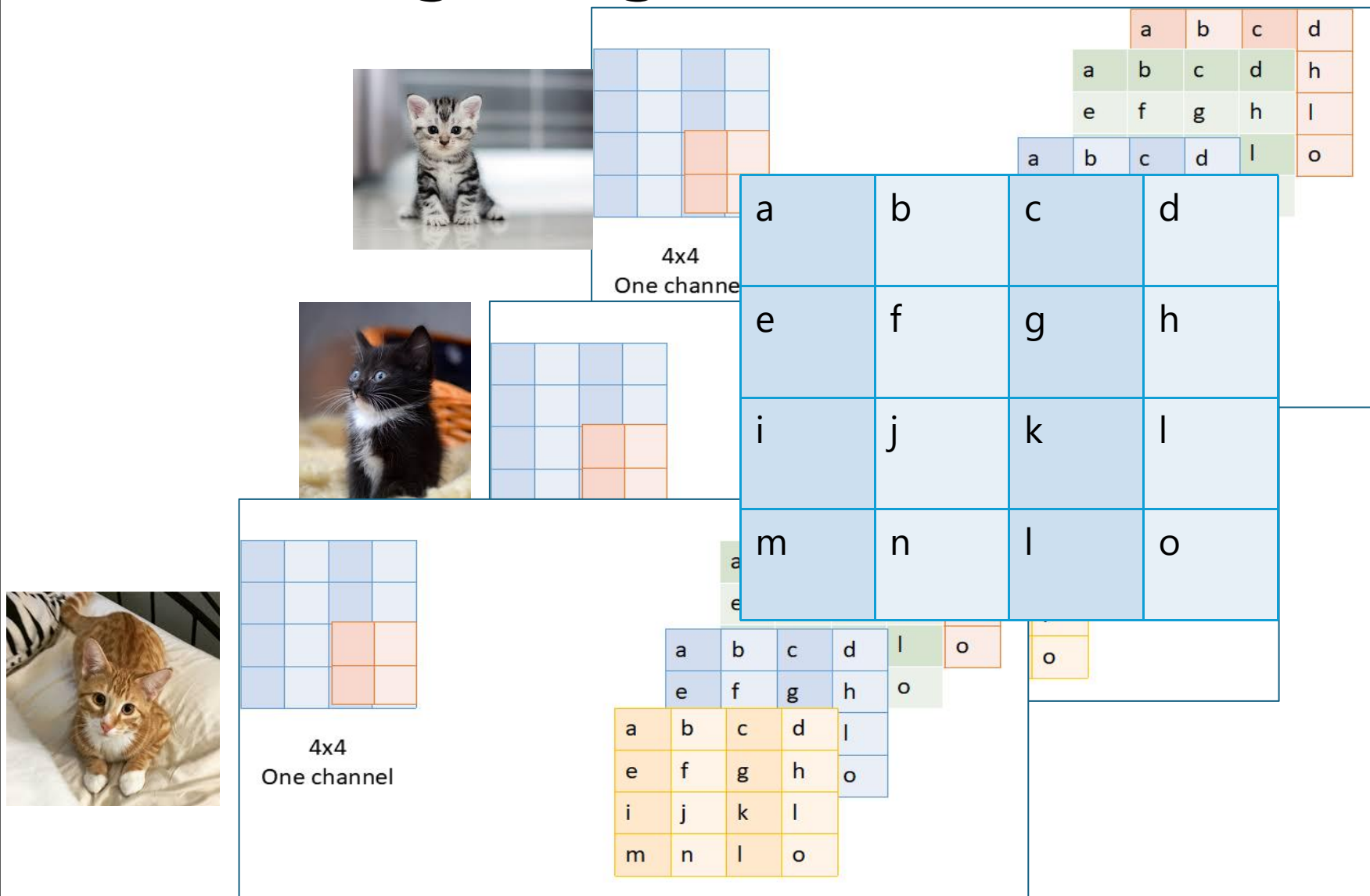
Pooling Stage



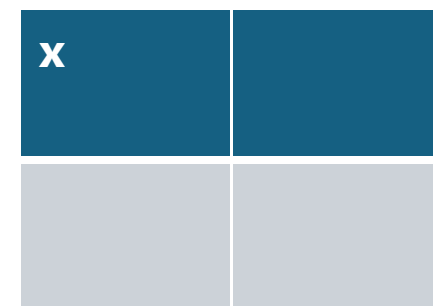
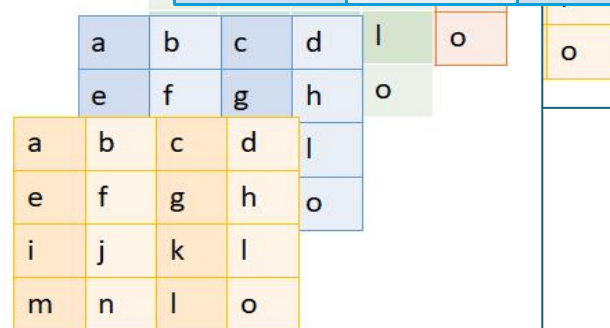
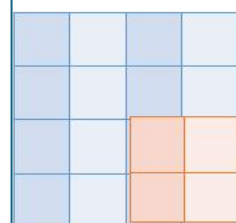
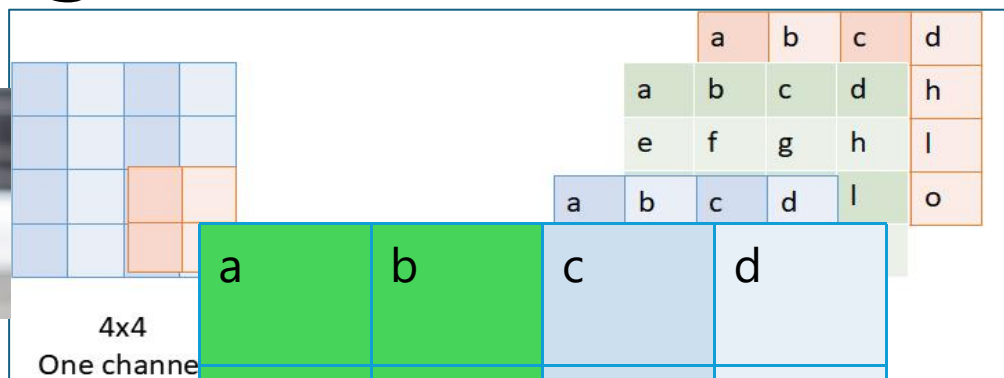
Pooling Stage



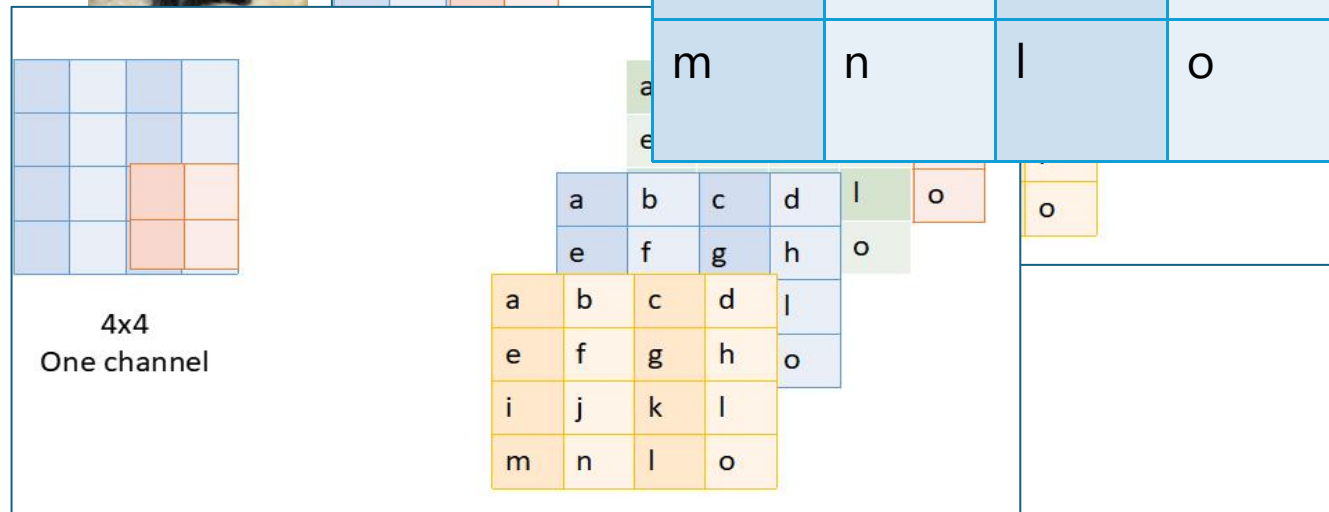
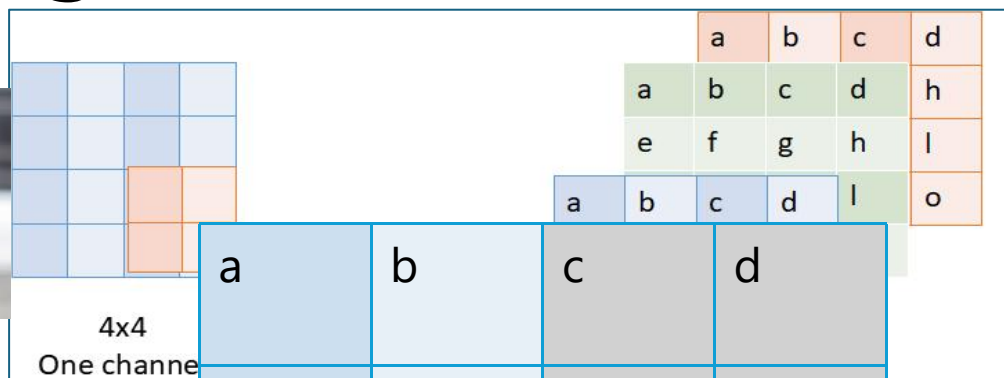
Pooling Stage



Pooling Stage



Pooling Stage



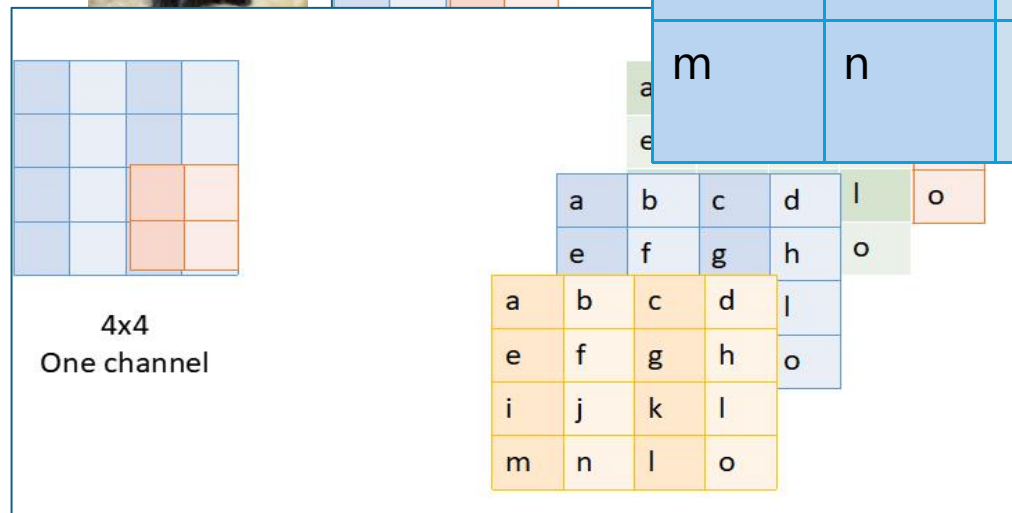
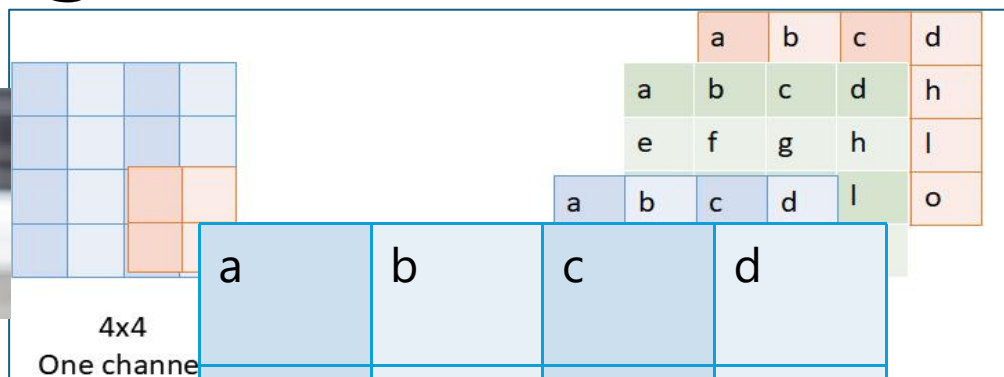
	a	b	c	d
a	b	c	d	h
e	f	g	h	l
a	b	c	d	l
				o

a	b	c	d
e	f	g	h
i	j	k	l
m	n	l	o

a	b	c	d	l	o
e	f	g	h	o	
a	b	c	d	l	
e	f	g	h	o	
i	j	k	l		
m	n	l	o		

x	y

Pooling Stage



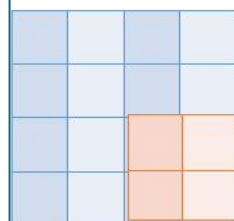
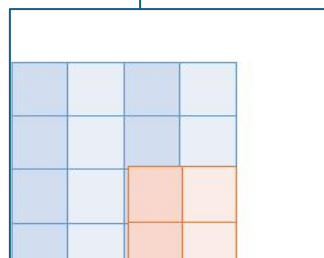
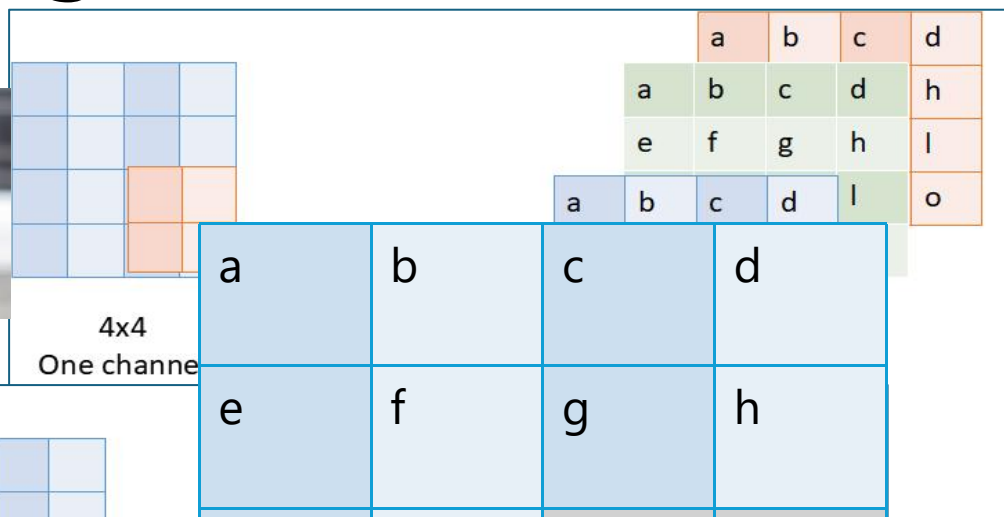
	a	b	c	d
a	b	c	d	h
e	f	g	h	l
a	b	c	d	l
				o

a	b	c	d
e	f	g	h
i	j	k	l
m	n	l	o

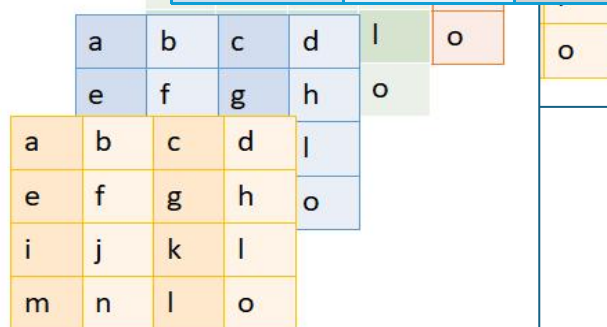
a	b	c	d	l	o
e	f	g	h	o	
a	b	c	d	l	
e	f	g	h	o	
i	j	k	l		
m	n	l	o		

x	y
z	

Pooling Stage

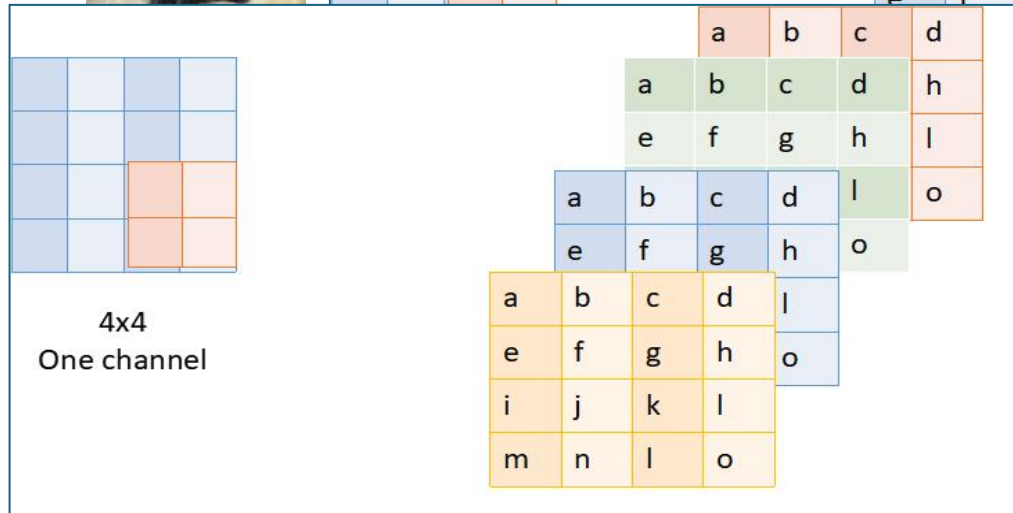
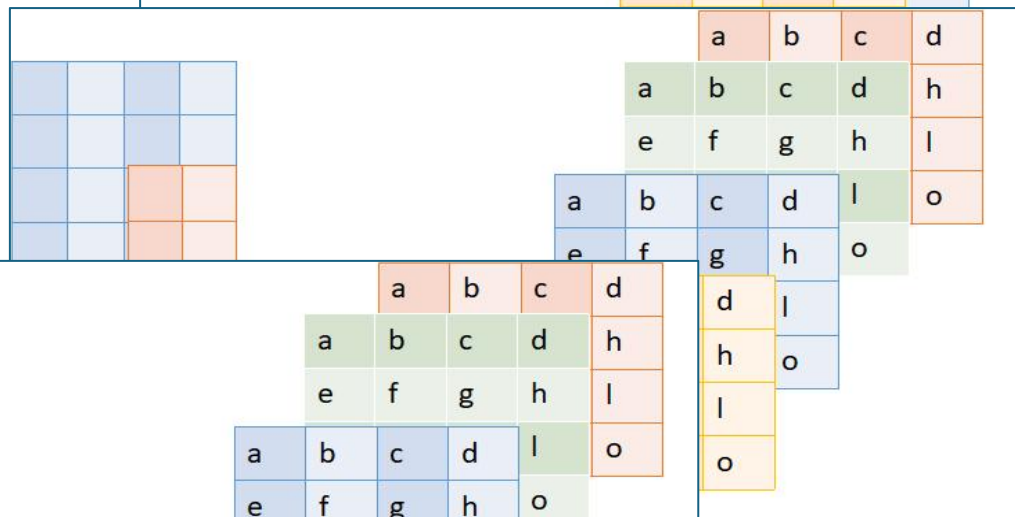
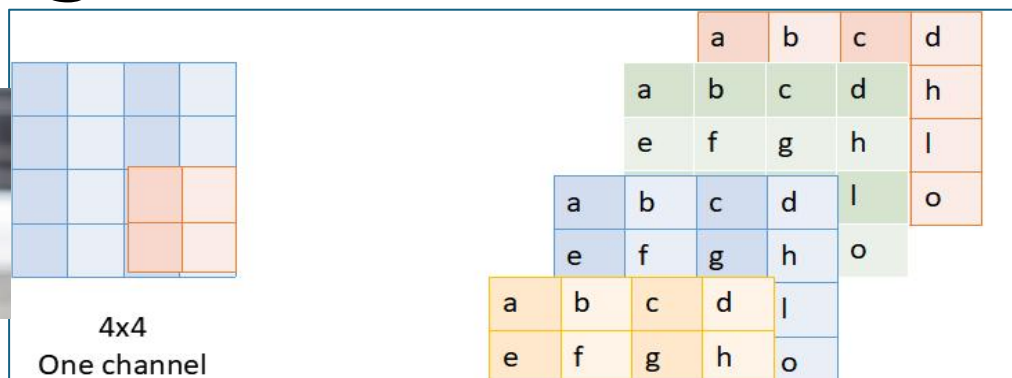


4x4
One channel



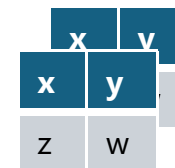
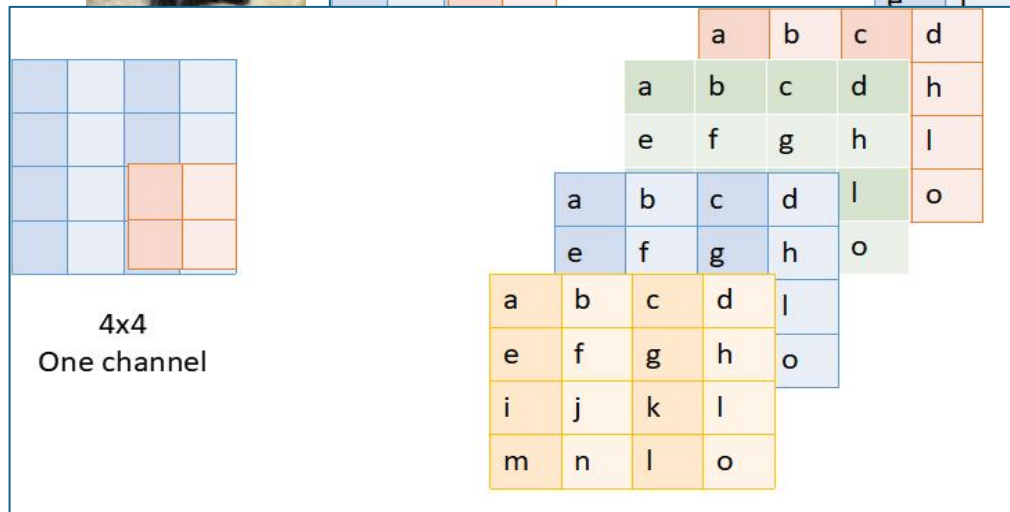
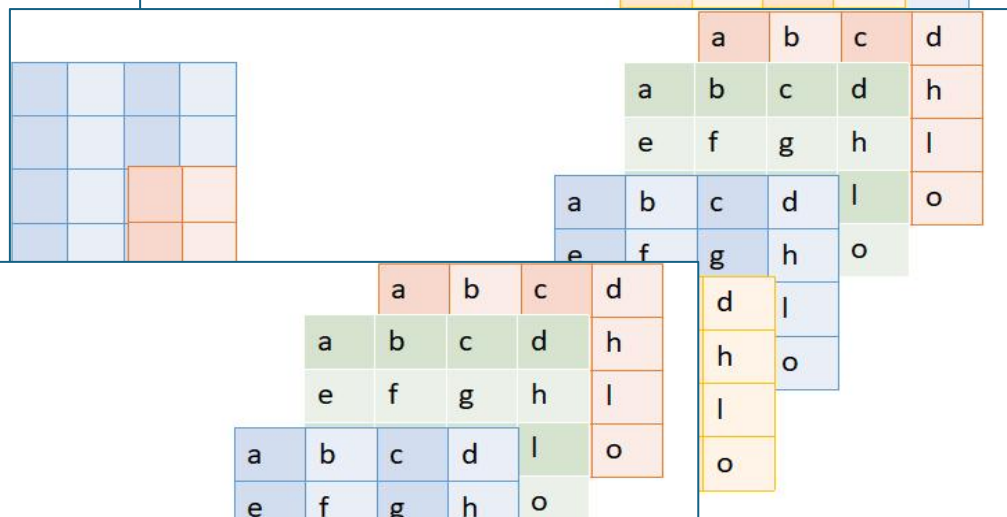
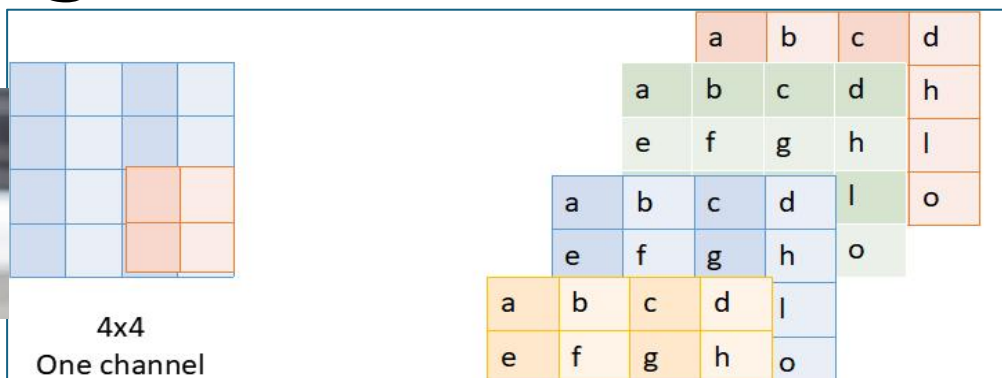
x	y
z	w

Pooling Stage

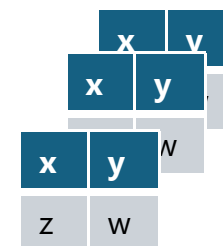
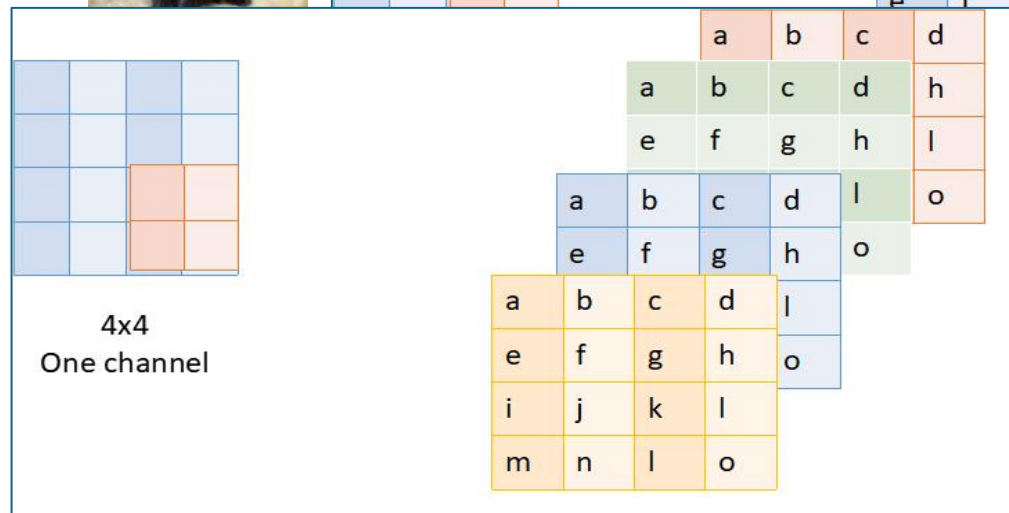
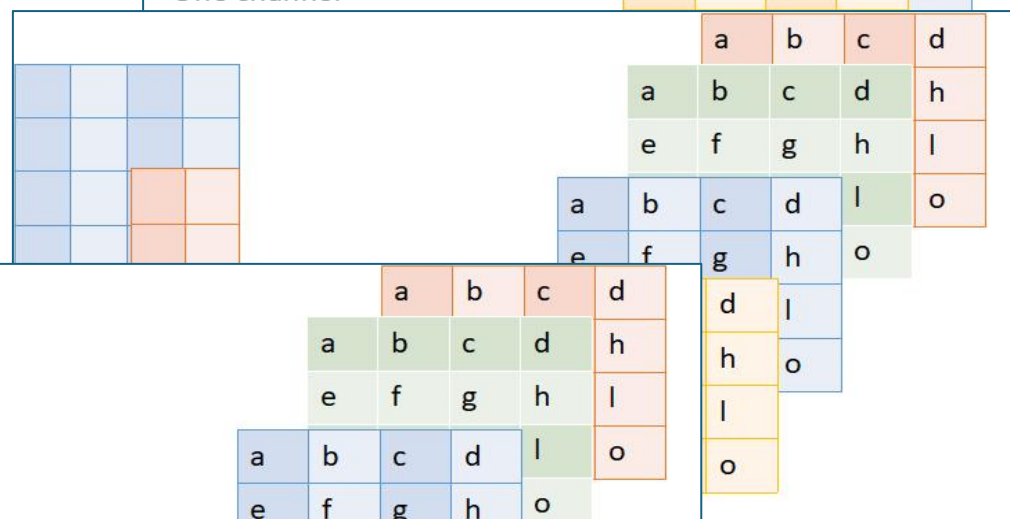
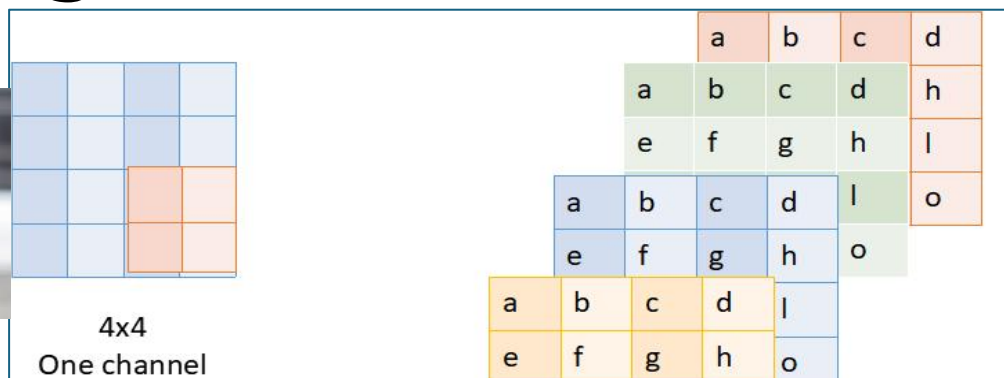


x	y
z	w

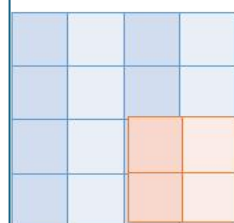
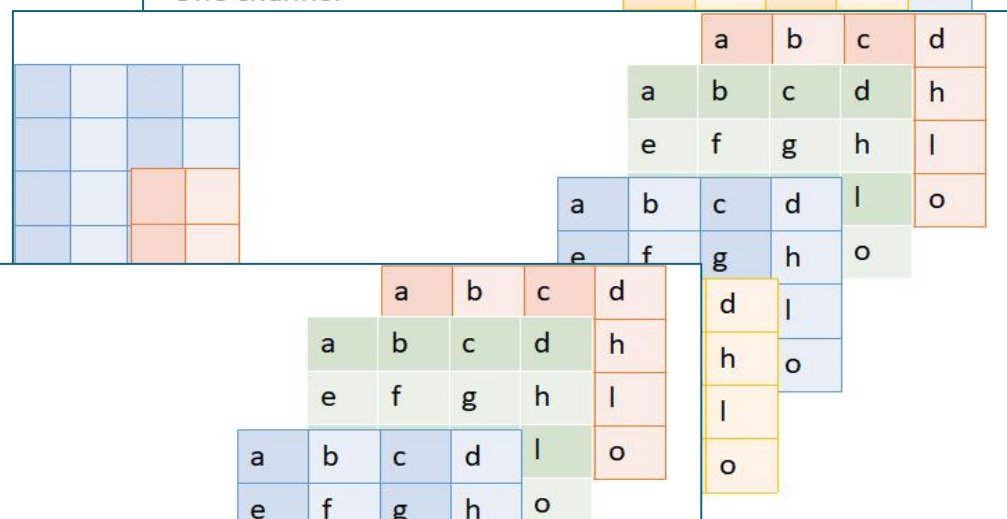
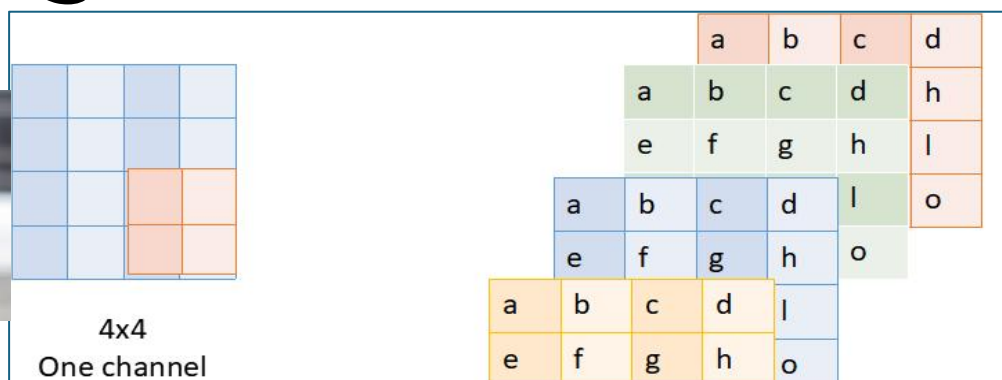
Pooling Stage



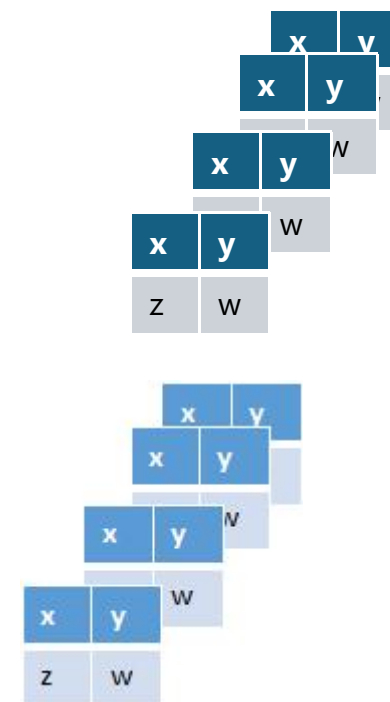
Pooling Stage



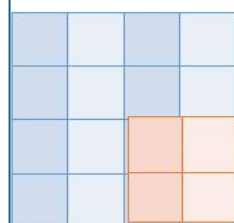
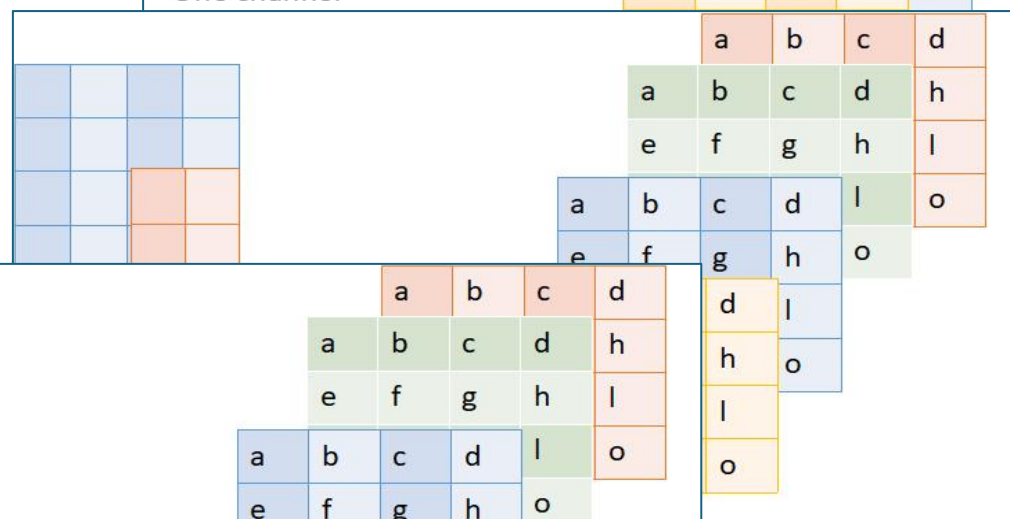
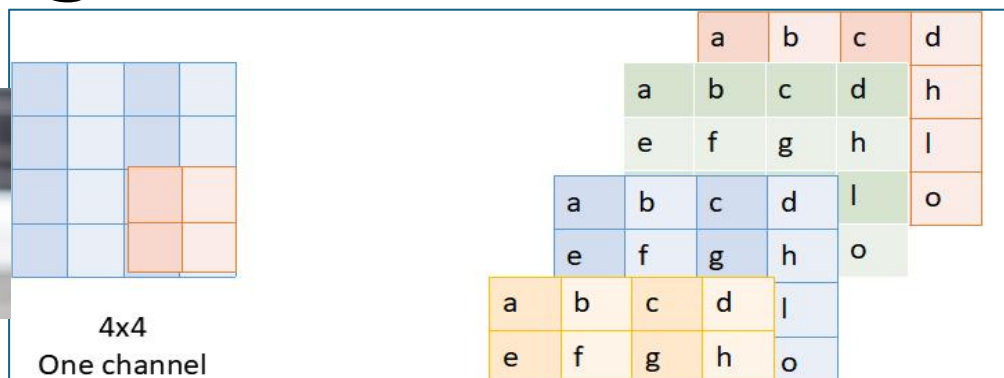
Pooling Stage



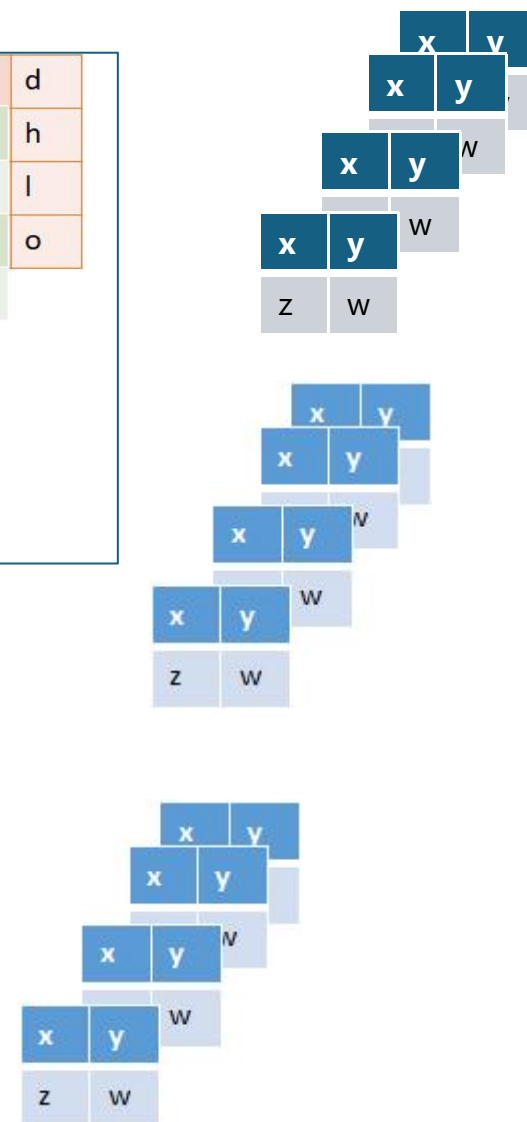
4x4
One channel



Pooling Stage



4x4
One channel



Summary Stages

